



Class - X

2023-24

Board: Maharashtra

Heredity and Evolution

- **Heredity:** The transfer of biological characters from one generation to the next one with the help of genes is called heredity.

Components of the DNA molecules: DNA molecule is made up of two helical strands consisting of deoxyribose sugar, phosphoric acid and pairs of nitrogenous bases.

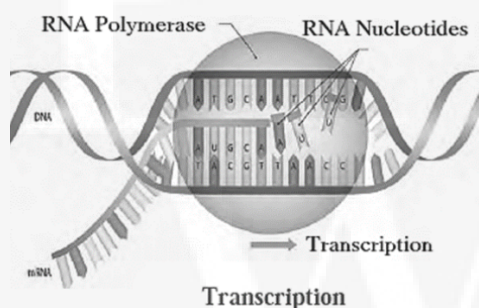
- **Central Dogma:**

Synthesis of proteins by DNA through the RNA is called central dogma.

- DNA Transcription → RNA Translation → Protein

- **Transcription, Translation and Translocation:**

Transcription: Synthesis of mRNA as per the nucleotide sequence present on the DNA molecule, is called the process of transcription.



- DNA → RNA

Triplet codon:

- The code for each amino acid consisting of three nucleotides, is called 'triplet codon'.

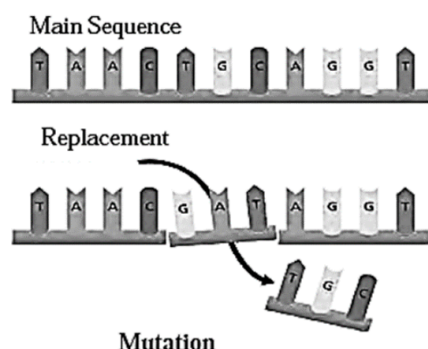
Translation: As per the codon on mRNA, the tRNA molecule with complementary 'anticodon' is brought near mRNA, this process is called 'translation'.

- RNA → Protein

Translocation: The ribosome keeps on moving from one end of mRNA to other end by the distance of one triplet codon. This process is 'translocation'.

- **Mutation:**

- Sudden change occurring in the genetic material is known as mutation.
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□ **Evolution:**

The gradual change occurring in living organisms over a long duration is called evolution.

(1) Simple elements → (2) Organic and inorganic compounds → (3) Complex Organic compounds such as proteins and nucleic acids → (4) Mixture of different types of organic and inorganic compounds. → (5) First primitive type of cells → (6) Processes to take up surrounding chemicals developed leading to growth of cells in numbers → (7) First living organism those organisms that could adjust and adapt to the surrounding conditions, survived and grew. This is according to the principle of natural selection. Some of them that could not adjust, perished.

□ **'Evidences of Evolution:**

(i) Morphological Evidences:

- Similarities in the external and visible features are called morphological evidences.
- These evidences shows different animal or plant groups must be the same having a common ancestor.

(ii) Anatomical Evidences:

- Human hand, cat's foreleg, flipper of whale and patagium of bat show difference in their features. .
- However, the structure of bones and bony joints in organs of each of those animals show similarity.
- This is an indication that all these animals originated from a common ancestor.

(iii) Vestigial Organs: The degenerated or underdeveloped organs which cannot perform any function are known as vestigial organs.

- Vestigial organs in human body are tail bone (coccyx), vermiform appendix, muscles of ear pinna, wisdom teeth, and body hairs.

(iv) Paleontological Evidences:

Fossil: Fossils are remnants of living organisms or their impressions which are preserved deep down in the earth's surface.

Carbon Dating: (Willard Libby) Carbon dating is a technique to understand the age of a fossil. When alive, animals and plants consume carbon continuously but this process ends after their death. Later their body carbon in the form of C-14 undergoes continuous decaying process. C-12 is not radioactive. Therefore, the ratio between C-14 and C-12 changes continuously.

(v) Connecting Links: Some organisms share morphological characters of two groups and hence they are called 'connecting links'.

Examples : (i) **Peripatus:** It shows characters of Annelida and Arthropoda

(ii) **Duck billed platypus:** Duck billed platypus is the connecting link between Reptiles and Mammals.

(iii) **Lung fish:** Lung fish is the connecting link between fishes and amphibians.



(vi) Embryological Evidences:

All the embryo of the vertebrates shows similar structure in initial developmental stage which shows that all of them have common origin.

Darwin's theory of natural selection:

- Theory of natural selection: 'The survival of fittest', i.e. organisms which are fit for survival, evolve while those that are not, perish. The natural selection thus acts to produce new species.
- Theory of natural selection is given in the book, 'Origin of Species' which is written by Darwin.

Important explanation of Theory of natural selection:

- All living organisms reproduce prolifically.
- There is competition with each other or struggle for survival.
- Organisms that show essential modifications for survival, sustain while remaining perish.

Objections to Darwin's theory:

- Some more factors are responsible for evolution and not just the natural selection.
- Any explanation about useful and useless modifications was not provided by Darwin.
- Causes of slow and abrupt changes was not explained by Darwin.

Lamarckism:

Lamarck proposed principle of 'use or disuse of organs and 'theory of inheritance/ancestry of acquired characters'. His concepts are Known as Lamarckism.

- Depending upon the activities or laziness of that organism these changes occur.

Examples of acquired characters:

- For several generations, giraffe extended the neck for browsing on leaves from heights. Due to such extension of neck, giraffe became long necked.
- Due to frequent hammering movements. Shoulders of the ironsmith became very, strong.
- Flightless birds like emu and ostrich have weak wings as they do not fly.
- Due to Wading and swimming in water the hind limbs of aquatic birds such as swan and duck became fitter for such mode.
- Due to burrowing habit, snakes lost the limbs.

Objections to Lamarckism:

- The view that unused organs degenerate and used ones evolve Was accepted but inheritance of such Characters was not agreed by the scientists,
- Modifications formed are not transferred to the next generation.

Speciation:

Species: The group of organisms that can produce fertile individuals through natural reproduction is called species.

Speciation depends upon following factors:

- Genetic variation
- Geographical changes
- Reproductive changes
- Geographical or reproductive isolation for a long period.

Human Evolution:

Dinosaurs
7 Crore Year
Lemur Ago
4 Crore Year
Egyptopithe Ago
cus



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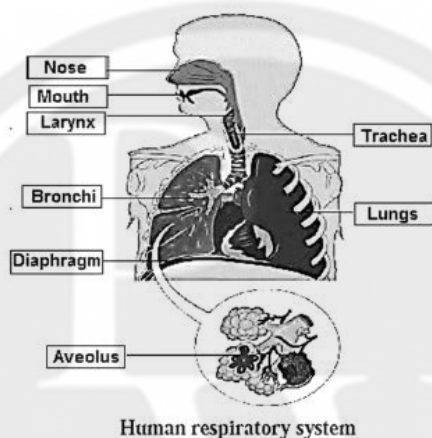
Life Processes in Living Organisms (Part-1)

Carbohydrates:

- **Source:** Milk, fruits, jaggery, cane sugar, cereals, vegetables, potatoes, sweet potatoes, sweet meats.
- **Functions:** Carbohydrates provide 4 Kcal energy per gram.

Respiration:

The process of breakdown of digested food leading to formation of energy (ATP) in the cell is called as respiration .



Living organism and Energy production:

In living organisms, respiration occurs at two levels as body and cellular level.

Body level respiration: (Breathing) Oxygen and carbon dioxide are exchanged between body and surrounding

Cellular level respiration: In this process, foodstuffs are oxidized either with or without help of oxygen inside the cell.

Cellular Level Respiration:

- Cellular respiration is done by following two methods, viz. aerobic respiration (In presence of oxygen) and anaerobic respiration (In absence of oxygen).

Aerobic respiration: The process of breakdown of digested food in the presence of oxygen is called as aerobic respiration. These are glycolysis, tricarboxylic acid cycle and electron transfer chain reaction.

(1) Glycolysis: (EMP pathway)

In glycolysis glucose molecule is oxidized stepwise into two molecules of each of pyruvic acid, ATP, NADH, and water.

- This process takes place in cytoplasm. Pyruvic acid formed during glycolysis is converted into a molecule of Acetyl Coenzyme-A and two molecules each of NADH₂ and CO₂.

(2) Tricarboxylic acid cycle:

- When Acetyl-Co-A molecules enter the mitochondria, the tricarboxylic acid cyclic chain reactions take place in mitochondria.
- Acetyl part of Acetyl-Co-A is completely oxidized releasing molecules of CO₂, H₂O, NADH₂, FADH₂.

(3) Electron transfer chain reaction:

- The electron transfer chain reaction takes place on inner membrane of mitochondria.



- From 1 NADH_2 molecule 3 molecules of ATP and from 1 FADH_2 molecule 2 molecules of ATP are produced during these cyclic reactions.
- Along with ATP, water molecules are also formed during chain reactions.

The two coenzymes that help in cellular respiration:

- NADH_2 - Nicotinamide Adenine Dinucleotide.
- FADH_2 - Flavin Adenine Dinucleotide.

□ **ATP:**

- ATP or Adenosine triphosphate is energy-rich molecule and energy is stored in the bonds by which phosphate groups are attached to each other.
- It contains a nitrogenous compound-adenine, pentose sugar- ribose and three phosphate groups.
- Total 38 molecules of ATP is obtained from one glucose molecule.

ANAEROBIC RESPIRATION PROCESS:

□ **Anaerobic Respiration:**

The process of breakdown of digested food in the absence of oxygen is called as aerobic respiration.

- Anaerobic respiration has two steps: Glycolysis and fermentation. In glycolysis glucose is incompletely oxidized releasing less amount of energy(2 ATP).
- Pyruvic acid produced through glycolysis is converted into other organic acids(Lactic acid fermentation) or alcohol (Ethanol) in process which is called fermentation.

□ **Energy from different food components:**

- Excess of the carbohydrates are stored in liver and muscles in the form of glycogen.

Proteins:

- Proteins are formed by amino acids which are held by peptide bonds.
- Proteins of animal origin are considered to be 'first class' proteins. Each gram of protein provides 4 Kcal of energy.
- Each gram of protein provides 4 Kcal of energy.
- excess of proteins is converted into glucose by process of gluconeogenesis

Lipids:

- Lipids are formed of fatty acids and alcohol (glycerol) which have specific bond between them. Fatty acids produce different substances in different cells.
- **Examples:** (1) Phospholipids - produce plasma membrane, (2) Progesterone, estrogen, testosterone, aldosterone etc. are hormones produced from fatty acids, (3) Covering around the axons of nerve cells.
- Each gram of lipids provides 9 Kcal of energy.

Vitamins:

- Vitamins are required for proper functioning and maintenance of the body.
- The main types of vitamins are Fat soluble vitamins: A, D, E and K, Water soluble vitamins: B and C.
- Riboflavin (Vitamin B2) and Nicotinamide (Vitamin B5) are necessary for their production of FADH_2 and NADH_2 respectively.

Water:

- Water is essential nutrient.

- Human body contains about 66 - 70% water. 70% water is present by weight in every cell. Blood- plasma has 90% of water.
- Loss of water or dehydration can cause problems with the functioning of cells and later that of the body.

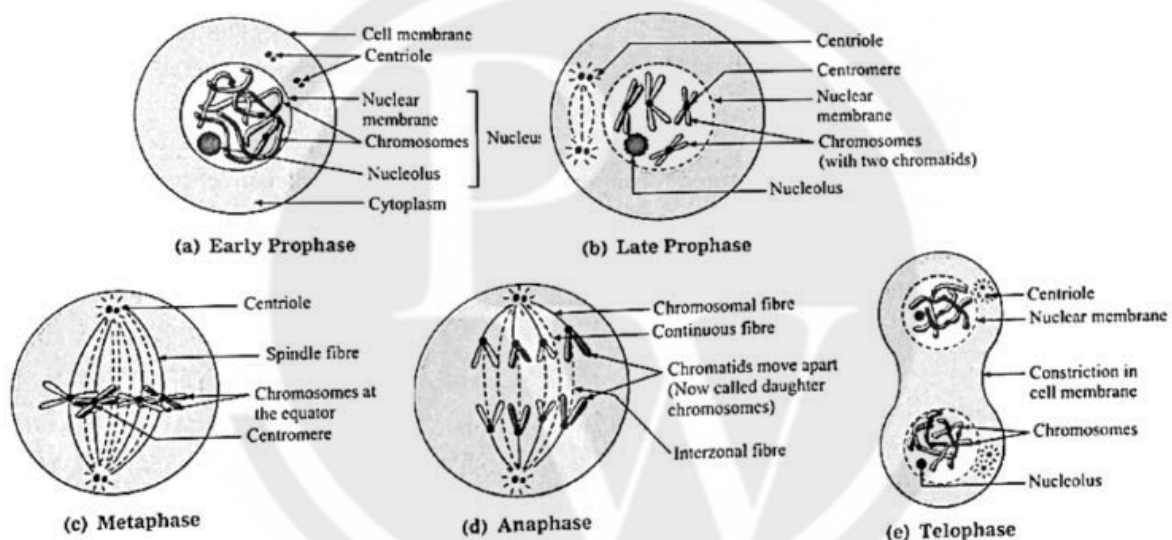
CELL DIVISION : AN ESSENTIAL LIFE PROCESSES:

Significance of cell division:

- A new organism is formed from existing one,
- A multicellular organism grows up
- Injured, emaciated body can be restored.

□ **Types of cell division:** (i) Mitosis and (ii) Meiosis

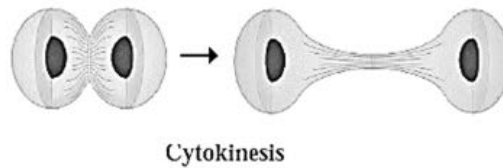
(i) Mitosis: There are two stages of mitosis. These are (a) Karyokinesis or nuclear division and (b) Cytokinesis or cytoplasmic division. Karyokinesis takes place in further four phases, viz prophase, metaphase, anaphase and telophase.



(a) Karyokinesis:

- **Prophase:** During prophase, condensation of chromosomes starts. The thin and thread like chromosomes start thickening. They are seen with their pair of sister chromatids. In animal cells the centrioles are seen to duplicate and move to opposite poles of the cell. Nuclear membrane and nucleolus disappear.
- **Metaphase:** Chromosomes complete their condensation and each one is seen with its sister chromatids. The chromosomes are seen in equatorial plane of the cell. The spindle fibres are formed from polar region, where centrioles are present, and they attach themselves to the centromere of each chromosome. Nuclear membrane now disappears completely.
- **Anaphase:** The centromeres of the chromosomes now divide forming two daughter chromosomes. The spindle fibres pull apart the chromosomes from equatorial region to the opposite poles. Chromosomes moving to the poles appear like bunch of bananas. One set of chromosomes reach each pole by the end of the anaphase.
- **Telophase:** Telophase is reverse of events that occurred in prophase. The thickened chromosomes decondense. They again assume the thin and thread like appearance. Nuclear membrane and nucleolus appear again. The spindle fibres are completely lost. The cell looks as if it has two nuclei in one cytoplasm.

(b) Cytokinesis: In animal cells a notch develops in the middle of the cell. This notch goes on deepening down and later the cytoplasm divides into two. In plant cells, cell plate formation takes place and then cytokinesis takes place.



Significance of mitosis:

- Essential for the growth of the body
- Necessary for restoration of emaciated body
- For wound healing
- For the formation of blood cells, etc.

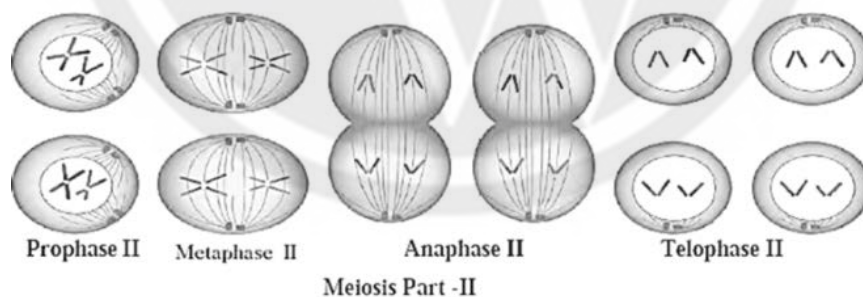
(ii) Meiosis:

There are two stages of meiosis: : (a) Meiosis-I and (b) Meiosis-II.

- By meiosis from one diploid cell, four haploid cells are formed.
- In meiosis crossing over between non-sister chromatids of homologous chromosomes occur. This results in genetic recombination.
- The four daughter cells formed after completion of meiosis are genetically recombined and not exactly alike and also not exactly similar to their parent cells.
- Spores and gametes are formed by meiosis.
- Meiosis helps to restore and maintain the chromosome number.

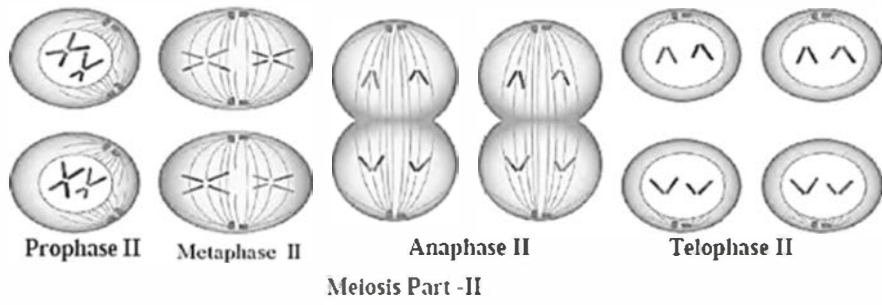
(a) Meiosis I:

- Homologous chromosomes undergo crossing over and hence there is genetic recombination.
- The homologous chromosomes are divided into two groups and from these, two haploid cells are formed.
- Prophase I of meiosis is lengthy phase which is subdivided into five phases, viz. leptotene, zygotene, pachytene, diplotene and diakinesis. During pachytene, crossing over takes place.



(b) Meiosis II:

- Meiosis II is similar to mitotic division. The two haploid cells that are produced in meiosis I now undergo further division forming four haploid.
- The diploid cells are $2n$ and the haploid are n . All the normal body cells are always $2n$ while only gametes formed by reduction division are n .
- In ' $2n$ ' condition, each type of chromosome is in pairs while in ' n ' condition, there is a single chromosome of each type.



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2023-24

CBSE : Grade 10
Class - X

Board: SSC

Life Processes in Living Organisms - 2

Submitted By : Mayur Nimbekar

□ **Reproduction: Asexual and Sexual:**

- Reproduction is formation of new offspring of same species by earlier existing parent organism.
- Reproduction is important character of living organisms.
- Evolution of every species occurs due to reproduction.

□ **Two main types of reproduction:** Asexual reproduction and sexual reproduction.

□ **Asexual reproduction:**

- Asexual reproduction is uniparental reproduction in which there is no formation of gametes.
- New organism is formed without the fusion of the gametes.
- The offspring produced is exactly similar to the parent organisms.
- It takes place by mitotic cell division.

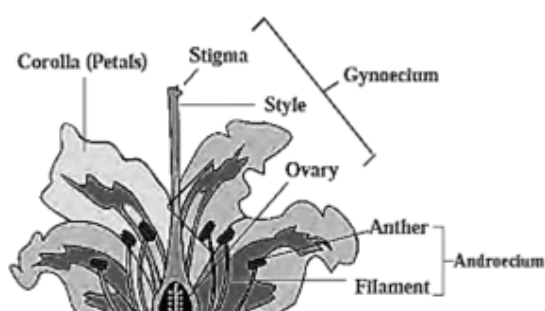
Demerit of asexual reproduction: Absence of genetic recombination.

Merit of asexual reproduction: Rapid process of reproduction

- **Asexual reproduction in unicellular organisms:** Binary fission(eg. Amoeba, Paramecium, Euglena), Multiple fission(Amoeba), Budding(Yeast)
- **Asexual reproduction in multicellular organisms:** Binary Fragmentation (eg. Spirogyra, sycon), Budding(Hydra), Vegetative propagation(stem of potato, root or carrot, etc), Spore formation (mucor)
- **Sexual Reproduction:**
 - Reproduction with the help of male and female gametes is called sexual reproduction. It has two main processes, viz. gamete formation and fertilization.
- **Gamete formation:** By meiosis the gametes are formed. The diploid germ cells give rise to haploid gametes.
- **Fertilization:** From union of haploid male and a female gamete a diploid zygote is formed during fertilization. Subsequent mitotic divisions of zygote form embryo which then develops into new individual.

Sexual reproduction in plants:

Flower is structural unit of sexual reproduction in plants.



these two are accessory floral whorls while two are essential floral whorls. They are protective in nature. They are usually green in colour. They protect the inner whorls. They can be of different colours.



- Androecium and gynoecium are essential whorls as they participate in sexual reproduction.
 - The male whorl androecium is made up of stamens. Each stamen has a filament with anther located at the upper end.
 - Gynoecium is made up of carpels, either in separate form or are united. Each carpel is formed of ovary at the basal end, hollow 'style' and the stigma at the tip of style. There are one or many ovules inside the ovary.
- **Bisexual flowers:** In bisexual flowers both androecium and gynoecium are located in the same flower. e.g. Hibiscus.
- **Unisexual flowers:** In unisexual flowers, androecium is present in male flowers and gynoecium is present in the female flowers, e.g. Papaya.

Pollination:

Pollination: Transfer of pollen grains from anther to stigma is called pollination.

Agents of pollination:

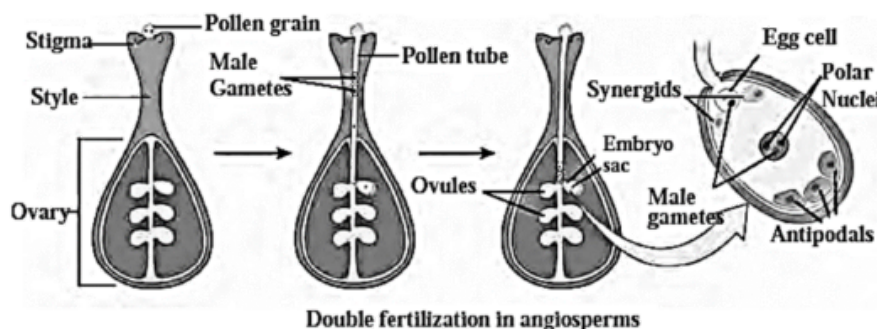
- **Abiotic agents:** Wind, water.
- **Biotic agents:** Insects, birds, or other animals.

□ **Types of pollination:**

- **Self – pollination:** Pollination involves only one flower, or two flowers borne on same plant.
- **Cross-pollination:** Involves two flowers borne on two plants of same species.
- In artificial pollination for forming new high yielding and resistant varieties of plants, the pollination with the help of brush, is done by scientists.

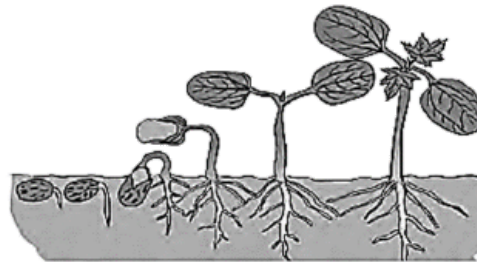
□ **Double fertilization:**

- Pollens fall upon sticky stigma and germinate.
- A long pollen tube and two male gametes are formed upon germination.
- The pollen tube carrying male gametes travels through style and reaches the embryo sac.
- In embryo sac, tip of the pollen tube bursts releasing two male gametes.
- Here fertilization occurs by union of one male gamete and egg cell.
- Second male gamete unites with two polar nuclei. This union forms endosperm.
- Since two nuclei take part in the process, it is called double fertilization.



□ **Germination:**

- The development of new plantlet from zygote after fertilization is called germination.



Seed germination

- After fertilization ovule develops into seed and ovary into fruit.
- Seeds from broken fruits fallen upon the ground start germinating if they get favourable conditions.
- Development takes place due to food stored in endosperm of seed.

□ **Sexual reproduction in human being:**

Human male reproductive system:

- It contains both the paired and unpaired structure.
- Paired: Testis, rete testes, vas eferens, epididymis, vas deferens, ejaculatory duct, seminal vesicles, cowper's gland
- Unpaired: urinogenital duct, prostate glands

□ **Human female reproductive system:**

- It contains both the paired and unpaired structure.
- Paired: ovary, oviduct, bulbo-urethral gland
- Unpaired: uterus, vagina

□ **Formation of gametes:**

- Sperm from father and ovum from mother are haploid gametes formed by meiosis.
- Man can produce sperms from puberty till death.
- But in a woman the function of reproductive system stops at menopause.
- In mature woman a single matured oocyte is released from ovary every month.
- In woman's body from birth, there are 2-4 million immature oocytes in the ovary of female foetus. Till the age of 45 years woman can produce ova.

□ **Fertilization:**

- Formation of zygote by union of sperm and ovum is called as fertilization.

Sex determination:

- Sex is determined according to the sex chromosomes,
- Human males have 44 somatic chromosomes and XY sex chromosomes.
- Human females have 44 somatic chromosomes and XX sex-chromosomes.
- Sperms produced are of two types viz. (22 + X) or (22 + Y) but ova/oocytes are all (22 + X) types.
- Type of sperm of father decides the sex of the child. If X bearing sperm fertilises the oocyte, the girl is born and if Y bearing sperm fertilises oocyte, it's a boy.

□ **Development and Birth:**

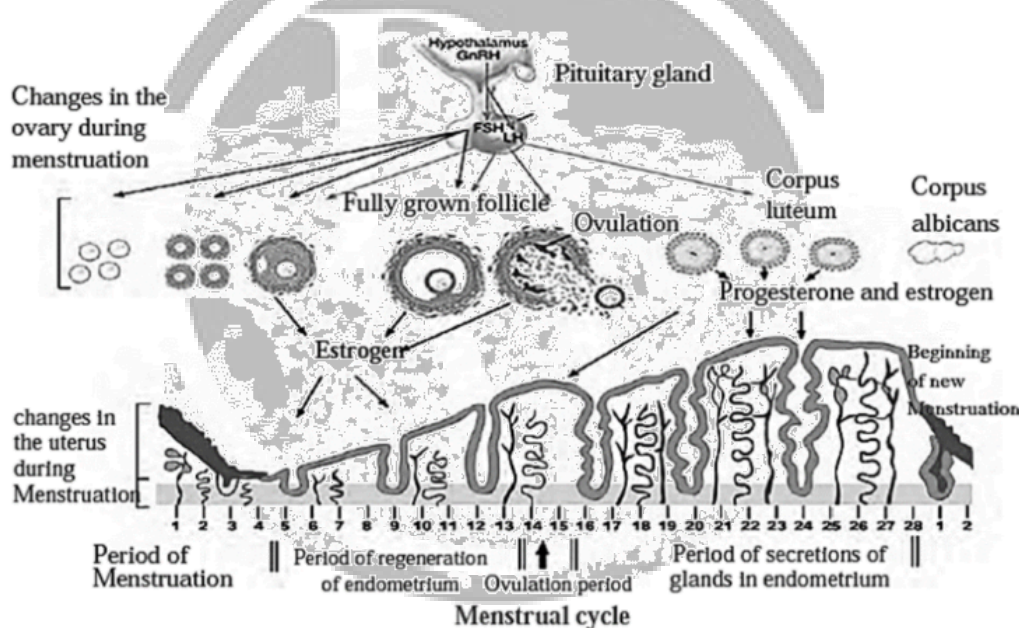
- The fertilization takes place in fallopian duct. The zygote thus formed undergoes rapid and repeated mitotic divisions to develop embryo.



- It travels from fallopian tubes to uterus. In uterus it gets implanted and grows for next 40 weeks or 9 months.
- The nutrition during this period is provided by placenta which is an organ developed in pregnancy mother.
- After completion of embryonic development for 9 months the pregnant mother gives birth to a baby.

□ **Menstrual Cycle:**

- Menstrual cycle is the naturally occurring repetitive changes in mature human female.
- These cyclic events are controlled by four hormones: (a) Follicle stimulating hormone (FSH) (b) Luteinizing hormone (LH) (c) Estrogen (d) Progesterone.
- FSH and LH are secreted from pituitary and estrogen and progesterone are secreted from ovary.
- One ovarian follicle develops along with the oocyte present in it due to effect of FSH. This developing follicle secretes estrogen.
- This follicle produces estrogen under the influence of FSH.
- Under the effect of estrogen, uterine endometrium develops or regenerates.
- The oocyte also undergoes development.
- Then under the influence of LH, ovulation takes place. Ovulation is bursting of ovarian follicle to release an oocyte.



- The remaining tissue of empty ovarian follicle forms a body called the corpus luteum. It is a secondary endocrine source and it starts producing progesterone.
- Under the influence of progesterone, the uterine endometrial glands secrete and make this endometrium ready for implantation of embryo.
- If oocyte is fertilized the endometrium forms placenta along with developing foetus.
- But if it is not fertilized, corpus luteum loses its function and becomes a degenerate body called corpus albicans.
- Corpus albicans does not secrete estrogen and progesterone.
- Due to this, endometrium degenerates and starts sloughing off.
- Degenerating endometrium, unfertilized ovum and blood is discarded out through vagina.
- This results into continuous bleeding for five days which is called menstruation.

□ **Modern reproductive technologies:**

- (i) **In Vitro Fertilization:** This is the technique in the modern medical field where childless couples can be blessed by their own child.
- IVF technique is used for childless couples who are faced with problems such as less sperm count, obstacles in oviduct, etc.



- The IVF technique is done by removing the oocyte from the mother and artificially fertilizing by the sperms collected from father. This fertilization is done in a test-tube. Thus, it is also called test tube baby. The embryo formed is implanted in uterus of real mother or a surrogate mother at appropriate time.

(ii) Surrogacy:

- In Woman if there is problem regarding the implantation of embryo in uterus, then help of another women is taken. This woman is called surrogate mother.
- Oocyte from real mother is taken out and fertilized with sperms collected from her husband. These gametes are fertilized outside in a test-tube and then the fertilized zygote is implanted in the surrogate mother.

(iii) Sperm Bank I (IVF) Semen Bank:

- If man has problems with the sperm production, then the sperms are collected from the sperm bank. Sperm bank is the place where the donors donate the sperms and such sperms are kept stored. The donor's identity is kept secret and he should also be physically and medically fit person.

□ **Twines:**

Twins are two embryos that develop simultaneously in the same uterus producing two offspring.

Two main types of twins are (i) Monozygotic twins (ii) Dizygotic twins.

- (i) **Monozygotic twins:** Formed from single embryo these twins are exactly alike and are of same gender. If within 8 days of zygote formation during the embryonic development cells of that embryo are divided into two groups, then monozygotic twins are formed.
- **Siamese twins:** In case of monozygotic twins, if the embryonic cells are divided into two groups 8 days after the zygote formation; there is high possibility of formation of conjoined twins (Siamese twins). Such twins are born with some parts of body joined to each other. Some organs are common in such twins.
- (ii) **Dizygotic twins:** When two oocytes are released from the ovary of woman and both are fertilized by two separate sperms then there is formation of dizygotic twins. These twins are formed due to two embryos that are separately implanted in the uterus. Such twins are genetically different and may be same or different by gender.

Reproductive health:

- The physical, mental and social well-being is called health.
- Syphilis and gonorrhoea are sexually transmitted or bacterial venereal diseases which affect people on a large scale.
- Symptoms of syphilis: Occurrence of chancre (patches) on various parts of body including genitals, rash, fever, inflammation of joints. alopecia, etc.
- Symptoms of gonorrhoea: Painful and burning sensation during urination, oozing of pus through penis and vagina, inflammation of urinary tract, anus, throat, eyes, etc.





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Board: Maharashtra

Environmental Management

Ecosystem: Biotic and abiotic components are presents in any environment. Ecosystem is formed by biotic and abiotic components and their interactions with each other.

□ **Components of Ecosystem:**

The different components in the ecosystem are as follows:

- **Abiotic components:** Air, water, soil, sunlight, temperature, humidity, etc.
- **Biotic components:** All the types of living organisms, like bacteria, fungi, plants and animals.

□ **Types of consumers:**

Primary, secondary, tertiary consumers or apex consumers are the different types of consumers in the ecosystem.

- These types are classified according to the trophic level to which they belong.
- **Food chain:** In every ecosystem, there are always interactions between producers, consumers and decomposers.
- This sequence of feeding interactions is called food chain.
- **Food web:** Food web is the collection of many small food chains. Food web is complex network of many small food chains. Thus, when many food chains are interlinked, they form food web.
- Different trophic levels in food chain:
 - Producers (First trophic level),
 - Primary consumers (Second trophic level),
 - Secondary consumers (Third trophic level),
 - Tertiary consumers (Fourth trophic level) etc.
- **Energy pyramid:** Energy pyramid is the diagrammatic representation, that depicts the energy levels at the various trophic levels.
- There are interactions in the form of energy transfer in all the food chains and food web. The energy pyramid shows how energy travels up a food chain.

Relationship between Environment and Ecosystem:

□ **Environment:** Everything that is around a living organism is called environment.

- Environment includes physical, chemical and other natural factors which surround the living organism.
- Many biotic, abiotic, natural and artificial factors together constitute conditions of the environment.

Two main types of environment:

- Natural environment: Air, atmosphere, water, land, living organisms, etc.
- Artificial environment: Artificial environment, directly or indirectly affects the natural environment.

Two main factors in environment: (1) Biotic factors (2) Abiotic factors.

- Continuous interactions occur between all the factors in the environment.



- **Ecology:** Ecology is the science that deals with the study of interactions between biotic and abiotic factors of the environment
- **Ecosystem:** Ecosystem is the basic functional unit used to study the ecology. Many ecosystems together form an environment.
- **Cycles operated in environment:** In environment there are 'bio-geo-chemical' cycles. These are of two types, viz. Gaseous cycles e.g. nitrogen cycle, oxygen cycle, etc. Sedimentary cycles e.g. phosphorus cycle.

ENVIRONMENTAL CONSERVATION:

Due to natural and man-made causes, there are many environmental problems on the earth. These problems affect the existence of various living organisms. In order to save these organisms and maintain the environmental balance, there is need for environmental conservation. Sensible or proper use of natural resources is a way of environmental conservation.

☐ **Natural Factors:**

- Among natural factors, the sudden changes in the weather, the different types of natural disasters etc. affect the normal environment. Due to such changes there are problems in the interrelationships that exist between food chain and food web.

☐ **Man-made factors:**

- Due to various man-made causes, there are extreme destruction of environment. Industrialization, the pollution due to such industries, Urbanization, hunting and poaching of wild animals, construction of dams, roads, bridges etc. are all man-made changes that cause lot of damage to environment.

☐ **Pollution:**

The environment is affected due to some natural factors of environment and some man-made factors such as pollution. Such factors create imbalance in the environment which in turn affect the existence of biota.

- **Environmental pollution:** Pollution brings about environmental degradation. It is largely due to natural or man-made causes. Pollution contaminates and makes the unnecessary and unacceptable changes in the surrounding environment. This causes direct or indirect changes in physical, chemical and biological properties of air, water and soil. These changes are usually harmful for all living beings, including human.
- **Reasons for pollution:** Population explosion, fast industrialization, and indiscriminate use of natural resources, deforestation, and unplanned urbanization are factors that pollute environment.
- **Our responsibility:** Pollution of air, water, noise, soil, thermal, light pollution etc. are different types of pollution that cause adverse effects. It ultimately affects existence of all the living organisms on the earth. Thus it is our responsibility to curtail polluting substances and aim at environmental conservation.
- **Need of environmental conservation:**
- Due to natural and man-made causes, there are many environmental problems on the earth. These problems affect the existence of various living organisms. In order to save these organisms and maintain the environmental balance, there is need for environmental conservation. If this is not done then there will not be any quality of life for the resident humans.
- **People's participation for environmental conservation:**
- Awareness about environment conservation rules and laws should be developed for common people.
- Large scale participation of the people in environment conservation
- Values like positive attitude towards environment and knowledge.
- Every nation has their own future plans about environmental protection and for this purpose they have formulated the laws.

☐ **Know This:**



Jadav Molai Payeng from Assam has made barren land into a forest which now stands on 1360 acres land. Now, this forest known as 'Molai Jungle'. It is rightly said that "Many people come together to destroy the forest, but a single person, if determined, can establish a new forest"! We must also have such a vision.

Environmental Conservation. Our social responsibility:

- Human - environment interrelationship existed since origin of man.
- Human being has become supreme on the earth due to his intelligence, memory, imaginations, creative ability etc.
- Human has used up natural resources without any thought. The development processes have caused extreme damage environment.
- Maintaining the environmental balance is the duty of humans.
- Since we have disturbed the environmental balance, only we should think of protection and conservation of nature.

Laws enacted about environmental conservation:

- Forest Conservation Act-1980 : The land reserved for forest conservation has been prohibited to use for any other pt this law.
- Ex. Permission of central government is compulsory for mining activities. Any person who disobeys this law is imprisonment for 15 days.
- Environmental Conservation Act-1986 : Purpose of this act is to control the pollution and punish the persons or harming the environment.
- Any person or factory is prohibited by this act from releasing the pollutants in atmosphere beyond a permissible limit. The person breaching this rule is entitled for either five year imprisonment or fine up to Rs. 1 lakh.
- **Wildlife Protection Act -1972:** Established for Protection of the wild life.
- Complete ban on trading of rare animals as per clause 49A
- Complete ban on use of articles prepared from skin or organs of wild animals as per clause 49B
- Compulsory disclosure of the stock of artefacts made from rare wild animals as per clause 49C

BIODIVERSITY:

The richness of living organisms in nature due to presence of varieties of organisms, ecosystems and genetic variations is called biodiversity. Biodiversity occurs at following three different levels.

☐ **Levels of biodiversity:**

Biodiversity is documented on the following three levels, viz. genetic diversity, species diversity and ecosystem diversity.

(i) Genetic Diversity: Diversity seen among the organisms of same species due to genetic differences is called genetic diversity. e.g. The individual human beings are different from each other. No two animals or plants are exactly alike.

(ii) Species Diversity: The difference between the different species is the species diversity. e.g. All the species of plants, animals and microbes which are seen in any natural environment.

(iii) Ecosystem Diversity: In one region there may be different ecosystems, such diversity in the ecosystems is called ecosystem diversity. Ecosystems are natural or artificial. Every region shows different types of ecosystems such as aquatic, terrestrial, desert or forest ecosystems. Each ecosystem has its own habitats with resident flora and fauna.

☐ **Sacred Groves:**

- A sacred grove is a green patch of forest that is protected by locals in the name of God.
- It does not belong to the Forest Service.
- Hunting of any kind is also prohibited here.
- In India, over 13000 sacred groves have been reported.



- Most of these are in Western Ghats in Maharashtra, Karnataka and Kerala. Also, in remaining parts of India sacred groves are reported.
- Role of sacred grove is tremendous in conserving the biodiversity..

How can biodiversity be conserved?

Hotspots of Biodiversity:

- Highly sensitive biodiversity spots in world : 34
- Areas of the Earth were occupied by these hotspots : 15.7 %
- Currently, sensitive areas that are destroyed : 86 %
- Presently left over sensitive spots on the earth : 2.3 %.
- Hotspots have 1,50,000 plant species which are 50% of the world count.
- In Eastern India jungles, 85 species are found out of 185 species of animals.
- In Western Ghats about 1,500 endemic plant species.
- Out of the total plant species in the entire world, 50,000 are endemic

Three Endangered Heritage Places of the Country:

Western Ghat: The Western Ghat spread over the states of Gujarat, Maharashtra, Goa, Tamil Nadu and Kerala has been endangered due to mining industry and search for natural gas. Habitats of Asiatic lion and wild bison of this region have been under threat.

Manas sanctuary: Manas sanctuary of the Assam is under threat due to dams and indiscriminate use of water. Tiger and rhino of that region are under threat.

Sunderban sanctuary: Sunderban sanctuary of West Bengal is reserved for tigers. However, the tiger population and overall local environment is seriously challenged by dams, deforestation, excessive fishing, trenches dug for same, etc.

Classification of Threatened Species:

- **Endangered Species:** Number of organisms is declined. Shrunk habitat which can lead to extinction. Ex. Lion tailed monkey, Lesser florican.
- **Rare Species:** Considerably declined number of these organisms. Endemic organisms may become extinct very fast. Ex. Red panda, Musk deer.
- **Vulnerable Species:** Extremely less number of organisms which further declines. Continuous declining number of organisms is a threat. Ex. Tiger, Lion.
- **Indeterminate Species:** Organisms appear to be endangered. The data is not enough as they have typical behavioural habits (like shyness). Ex. Giant squirrel (Shekhru).





Class - X

2023-24

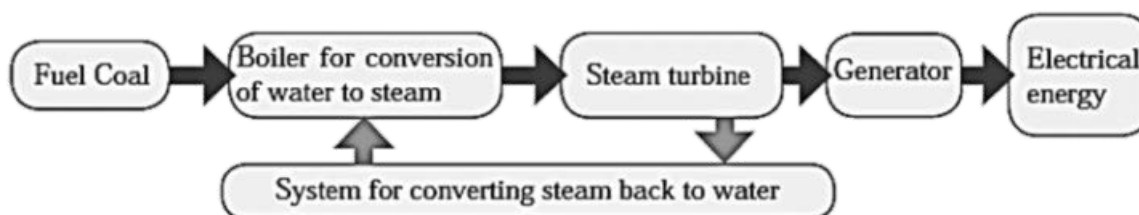
Board: Maharashtra

Towards Green Energy

- **Energy :** The capacity to do work is called energy.
- **Different forms of energy :** Heat energy, light energy, electric energy, solar energy, chemical energy, nuclear energy, mechanical energy, etc. are such forms of energy which we use to fulfil our requirements.
- **Electric Energy :**
 - Electric energy is used in a variety of daily items, including lamps and streetlights. Fans, an iron, a refrigerator, a water pump, a phone charger, a TV, a tape recorder, and a computer.
- **Generation of Electric energy :**
 - Michael Faraday invented the principle of electromagnetic induction. This principle is largely used in generation of electricity.
 - This principle is as follows: A potential difference is generated across the conductor whenever magnetic field around a conductor changes.
 - In two ways the field around a conductor is changed. (i) Keeping the conductor stationary and rotating the magnet. (ii) Keeping the magnet stationary and changing the field around the conductor.
 - In commercial power generation plants Large Generator rotate by turbines. Turbine having blades rotates the magnet in the electric generator. Due to rotation of magnet potential difference is generated across the conductor whenever magnetic field around a conductor changes, results in production of Electric energy.
- (1) **Thermal energy based electric power station :**
 - In thermal power plant the turbines are rotated using steam. Here the coal is burnt.
 - The heat energy liberated from this burning is used to heat the water in the boiler. This water produces steam of very high temperature and pressure.
 - The kinetic energy in the steam rotates the turbines.

Energy transformation in thermal power plant

Chemical energy in coal → thermal energy → kinetic energy in steam → kinetic energy in turbine
→ Generator → electric energy



Flow chart showing generation of electrical energy using thermal energy

- **Problems caused due to thermal power plants :**
 - Burning of coal results into air pollution .
 - Serious health problems arise due to toxic gases and soot particles .
 - The coal reserves are limited.



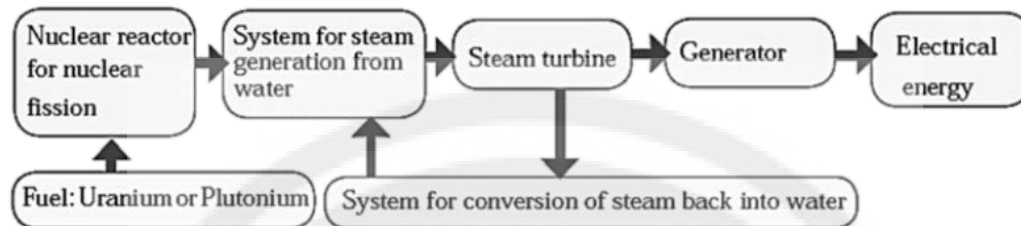
(2) Power plant based on Nuclear Energy :

- In Nuclear Power plant, energy released by fission of nuclei of Uranium or Plutonium atoms is used to generate the steam of high temperature and high pressure.

Energy transformation in nuclear power plant

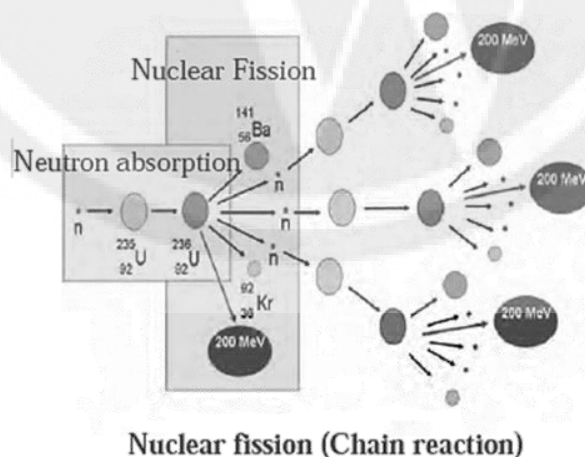
- Energy in the atoms → nuclear energy → converted into thermal energy → converted into kinetic energy of steam → converted into kinetic energy of turbine → converted into electrical energy.

Flow chart of nuclear power plant :



Formation of nuclear energy :

- Upon bombardment by neutrons on atom of Uranium-235, absorption of neutron takes place. It converts Uranium-235 into its isotope Uranium-236,
- Being extremely unstable, Uranium-236 transforms into Barium and Krypton atoms through a process of fission. This releases three neutrons and 200MeV energy.
- Three more Uranium-235 atoms undergo fission due to three neutrons generated in this process, Thus, more fission releases more energy,
- Thus, a chain reaction takes place in which the process of fission of Uranium -235 atoms continues,
- In this way a controlled chain reaction is carried out to release thermal energy, which in turn is used for generating electric energy.



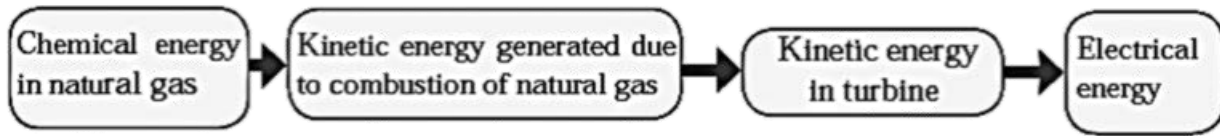
Disadvantages of nuclear energy :

- Radioactive and harmful radiations are emitted after fission of nuclei occurs.
- The nuclear waste of radioactive substance requires very safe disposal.
- Very fatal accidents can happen in power plant which emits very harmful radiations.

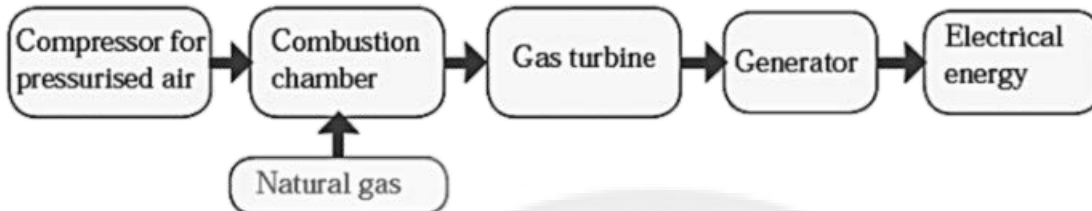
(3) Power generation plant based on energy of natural gas:

- Combustion of natural gas generates very high temperature and pressure. This gas is used to run the turbine.

Step-by-step transformation of energy in this plant :



Arrangement in power plant based on energy of natural gas :



- Into the combustion chamber pressurized air is introduced by a compressor → the natural gas burns due to air → high temperature and pressure → runs the turbine → Turbine drives generator → Electricity is produced.

Advantages of Using of natural gas in thermal power plant :

- More efficiency than coal power plants
- Less polluting as there is no sulphur in natural gas.
- Less pollution by burning of natural gas.

Electric energy generation and environment

Problems :

- Incomplete combustion of fuels releases Carbon monoxide which is harmful and gives adverse effects on health
- Burning of fossil fuels like coal, and natural gas, emitted toxic gases and soot Particles in air due to this air pollution increases causes, problems in respiratory system, like asthma.
- Due to burning of fuels, increase in percentage of carbon dioxide in the air, Global warming and climate change seen.
- Burning of fuels like coal, diesel, petrol, etc. releases Nitrogen dioxide cause acid-rain.
- The coal reserves in the world are thought to last in next 200 years and the natural gas reserves for about 200-300 years.

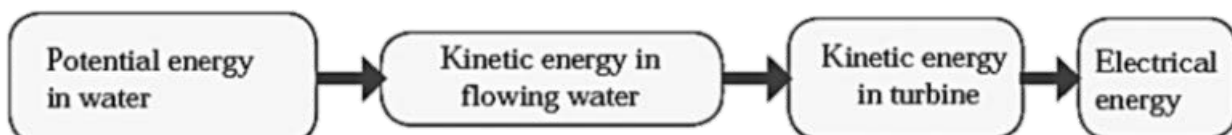
(4) Hydroelectric power plant:

- In hydroelectric power plant, the potential energy in water stored in dam is converted into kinetic energy of water.

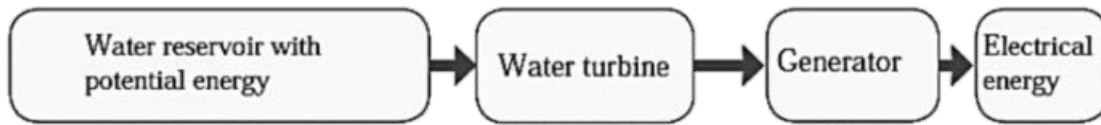
Process of Hydro electric energy generation :

Hydroelectric power plant → Stored water's potential energy → Fast flowing water → Converted into kinetic energy of water → Such speedy water from dam is brought. → to the turbine at the bottom level → Kinetic energy of the flowing water → drives the turbine → turbine then drives the generator → electricity is generated.

Energy Transformation in hydroelectric power plant :



Different Stages in hydroelectric power plant :



Advantages of hydroelectric power generation :

- Since there is no combustion, there is no pollution.
- Possible to generate electricity as and when required.
- Water supply can also be achieved due to dam.
- Rains replenish the stored reservoir of the water.

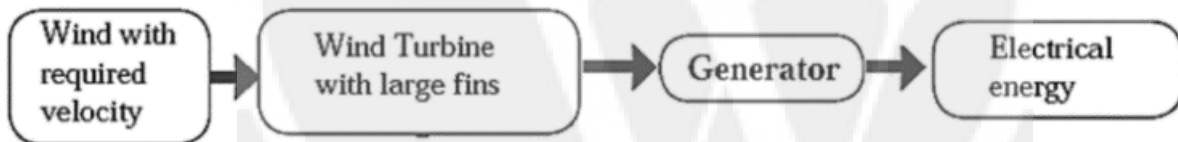
Disadvantages of hydroelectric power generation :

- Some villages or settlements are submerged.
- People are displaced and their rehabilitation is the major problem.
- Fertile land, agricultural plots forests and biodiversity is lost forever.
- Obstruction of the flowing water leads to adverse effect on the resident aquatic organisms.

Electricity generation using wind-energy :

- The kinetic energy of wind can be converted to electrical energy with the help of wind-turbine.
- By using wind turbines can produce electricity in the range less than 1 kW to about 7MW (7000 kW).

Stages in electric generator using wind energy :



Transformation of energy in an electric generator using wind energy :



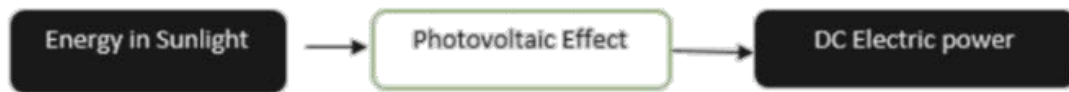
Electric Energy generation using solar energy :

Solar energy can be used to make electric energy in following two ways :

- Using radiations from the sunlight, without using a transformer, it is trapped directly into solar photovoltaic cells. Electrical energy is produced from solar energy due to direct conversion occurring in photovoltaic cells.
- Solar energy is converted into thermal energy. By using this thermal energy turbine-generator is rotated and by that thermal energy electricity is generated.

Solar photovoltaic cell :

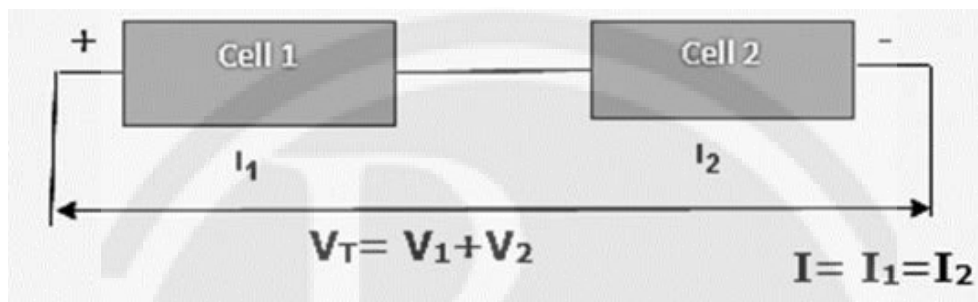
- Solar photovoltaic effect is the conversion of solar energy directly into electrical energy.
- Such generated energy is DC in nature.
- A special type of material called semiconductor (e.g. silicon) is used in making solar cells.
- The potential difference available from a solar cell is independent of its area.



Connections of solar cells :

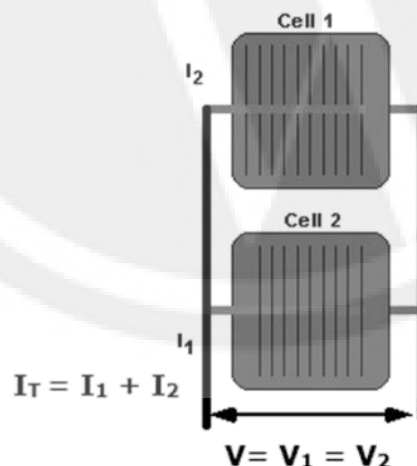
(a) Solar cells connected in series:

- When two solar cells are connected in series, the potential difference obtained from both the cells is sum of the potential differences of individual solar cells ($V = V_1 + V_2 + \dots + V_n$).
- When solar cells are connected in series, currents from the individual cells cannot be added ($I = I_1 = I_2 = I_n$).



(b) Solar cells connected in Parallel:

- When two solar cells are connected in parallel, the potential difference obtained from this combination is the same as the potential difference obtained from individual cell ($V = V_1 = V_2 = V_n$).
- The current generated from two cells in parallel is the summation of the currents from an individual cell ($I = I_1 + I_2 + \dots + I_n$).



Solar panel :

- The basic unit in solar electric plant → Solar cell → Solar panel → By joining in a series → String → Many strings joined in parallels → Solar array.
- Calculators running on solar energy need marginal power, whereas power stations need power in MW capacity.
- The DC power is available from the solar cells.
- Therefore, it can be used in applications like electric LED (Light Emitting Diodes) and other appliances that work on DC power.
- DC solar power is converted to AC power with the help of an inverter.



Advantages of solar energy :

- Generation of energy without any fuel combustion and without any pollution.

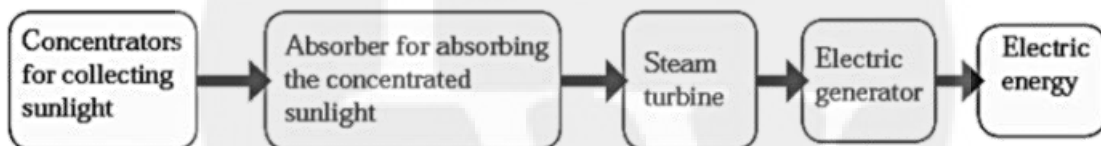
Problem of solar energy :

- Available only on sunny days and during day-time only.

Solar Thermal power plant :

- This solar thermal energy is used for electricity production.
- Solar radiation is absorbed by many reflectors that reflect and concentrate it.
- This solar energy is converted into heat energy which is then used to make steam which drives turbines and generate electricity through generators.
- Different stages in solar thermal power plant :

Different stages in solar thermal power plant :





Class - X

2023-24

Board: Maharashtra

Animal Classification

History of Classification:

- Scientists:** Greek philosopher Aristotle Theophrastus, Pliny, John Ray, Linnaeus.
Basis for classification: Body size, habits and habitats
Method: Artificial method
- Scientists:** Dobzhansky and Mayor
Basis for classification: Body organization, types of cells, chromosomes, bio-chemical properties
Method: Natural system of classification
- Scientists:** Carl Woese
Basis for classification: Presence or absence of notochord.
Method: Method based on evolution Traditional method.

Non-Chordates:

- Notochord: Absent
- Pharyngeal Gill slits: Absent
- Nerve cord: Ventral, solid and paired
- Heart: Dorsally locate

Chordates:

1. Presence of notochord in the body either throughout the life or at least in the developmental stage.
2. Presence of pharyngeal gill slits. (at least during developmental stage).
3. Presence of single, tubular and dorsal spinal cord.
4. Presence of ventrally situated heart.

Robert Whittaker has given five kingdom classification system. All the multicellular, heterotrophic organisms are included under kingdom Animalia or Animal kingdom.

According to this system the criteria for classification are as follows:

(1) Body organization (2) Body symmetry (3) Body cavity (4) Germinal layers (5) Segmentation.

New system of animal classification:

Criteria for new system of classification:

- Grades of organization:** Unicellular organisms have a single cell. But in multicellular animals there are many cells.

Grades of Organization	
Protoplasmic grade	Body made up of only protoplasm. E.g., Unicellular animals.
Cellular grade	Many cells are present. But tissues are not formed. E.g., Porifera
Cell-tissue grade	Cells performing similar function form the tissues. Tissues perform all body functions. E.g., Cnidaria
Tissue Organ grade	Tissues are organized to form some organs. But complete organ systems are not formed. E.g., Flat worms
Organ-system grade	Different organs form organ system to perform specific functions in the body. E.g., Human beings and other higher animals

2. Body Symmetry:



If body of any animal is cut through imaginary axis of body, it may or may not produce two equal halves.

Asymmetrical Body: In case of such body, there is no any such imaginary axis of the body through which we can get two equal halves. Ex. Amoeba, Paramoecium, some sponges.

Radial Symmetry: In this type of body, if imaginary cut passes through central axis but any plane of body, it gives two equal halves. Ex. Star fish.

Bilateral Symmetry: In this type of body, there is only one such imaginary axis of body through which we can get two equal halves. Ex. Insects, fishes, frog, birds, human, etc.

3. Germ Layers:

- During the initial embryonic period of any multicellular animal there is formation of germinal layers or germ layer.
- The primitive animals have only two germ layers called ectoderm and endoderm. This condition is called diploblastic.
- The higher animals have three germ layers: ectoderm, mesoderm and endoderm. This is called triploblastic nature.

4. Body cavity (Coelom):

Cavity between the body and internal organs is called as body cavity/coelom.

Eucoelomate: Body cavity present. During initial embryonic development period body cavity is formed from either mesoderm or gut. Example- All animal phyla from Annelida to higher phyla chordata .

Acoelomate: Absence of coelom. Example- Porifera, Cnidaria and Platyhelminthes.

Pseudocoelomate: Body cavity is present, but it is not formed by the mesoderm or gut. Hence it is not true coelom. Example- Aschelminthes.

5. Segmentation: Segmentation is the phenomenon in which the body of animals is divided into segments E.g., Phylum Annelida and Arthropoda.

1) Phylum- Porifera

- **Body Structure:** These animals are with simplest body plan and are called as 'Sponges'. They bear numerous pores on their body. Those pores are called as 'Ostia' and 'Oscula'.
- **Habitat:** These are aquatic animals. Most of them are marine and few are freshwater dwellers.
- **Body Symmetry:** Most of the animals have asymmetrical body.
- **Characteristics:** These animals have special types of cells- collar cells.
- Their spongy body is supported by spicules or sponging fibres. Spicules are made up of calcium carbonate or silica.
- These animals feed upon small organisms taken in their body along with water. Water is taken in through ostia and given out through oscula.
- Germinal Layers coelom: Acoelomate, cellular grade
- **Reproduction:** These animals reproduce by budding, an asexual method and / or by sexual method. Besides, they have good ability of regeneration.
- **Examples:** Sycon, Euspongia (Bath sponge), Hyalonema, Euplectella, etc.

3) Phylum - Coelenterata/Cnidaria

- **Body Structure:** Body of these animals is cylindrical or umbrella-like. If it is cylindrical, it is called as 'Polyp' and if it is umbrella like called as 'Medusa'.
- **Habitat:** Most of these animals are marine. Only few are fresh-water dwellers.
- **Body Symmetry:** Body of these animals is radially symmetrical and diploblastic.
- **Characteristics:** Cnidoblast bearing tentacles are present around the mouth. Tentacles are useful for capturing the prey whereas cnidoblasts inject the toxin in the body of prey. Those are useful for protection too.
- **Locomotion:** sedentary or free swimming
- **Germinal Layers coelom:** Acoelomate, Diploblastic.



- **Reproduction:** Asexual, Sexual
- **Examples:** Hydra, Adamsia (Sea anemone), Physalia (Portuguese- man-of war), Aurelia (Jelly fish), Corals, etc.

4) **Phylum - Platyhelminthes**

- **Body Structure:** Body of these animals is slender and flat like a leaf or strip. Hence, they are called as 'flatworms'.
- **Habitat:** Most of these animals are endoparasites, Few are free-living and aquatic.
- **Body Symmetry:** Bilaterally symmetrical.
- **Characteristics:** Flat body is a typical character.
- **Germinal Layers coelom:** Body is Acoelomate and triploblastic i.e. their body is made up of three germ layers- endoderm, ectoderm and mesoderm.
- **Reproduction:** Bisexual, Sexual. These animals are hermaphrodite i.e. male and female reproductive systems are present in the same animal body.
- **Examples:** Planaria, Liver fluke, Tapeworm, etc.

5) **Phylum- Aschelminthes**

- **Body Structure:** Body of these animals is long thread-like or cylindrical. Hence, they are called as round worms.
- **Habitat:** These animals are either free living or endoparasites. Free living animals are either aquatic or terrestrial.
- **Body Symmetry:** Bilaterally symmetrical.
- **Germinal Layers coelom:** Body of these animals is triploblastic and pseudocoelomate.
- **Characteristics:** Body of these animals is non-segmented and covered with tough cuticle.
- **Reproduction:** These animals are unisexual.
- **Examples:** Ascaris (Intestinal worm), Filarial worm, Loa loa (Eye worm), etc.

6) **Phylum - Annelida**

- **Body Structure:** Body of these animals is long, cylindrical and metamerically segmented.
- **Habitat:** Most of the animals are free-living, but few are ectoparasites. Free-living animals may be marine or freshwater dwellers or terrestrial.
- **Body Symmetry:** Bilaterally symmetrical
- **Germinal Layers Coelom:** These animals are triploblastic, and eucoelomate.
- **Locomotion:** They have setae or parapodia or suckers for locomotion.
- **Characteristics:** Their body is covered with special cuticle.
- **Reproduction:** These animals are either hermaphrodite or unisexual.
- **Examples:** Earthworm, Leech, Nereis, etc.

7) **Phylum- Arthropoda**

- **Body Structure:** These animals have jointed appendages. Hence, they are called as arthropods.
- **Habitat:** These animals are found in all types of habitats ranging from deepest oceans to highest mountains.
- **Germinal Layers coelom:** Body of these animals is triploblastic, eucoelomate.
- **Body Symmetry:** Bilaterally symmetrical and segmented.
- **Characteristics:** Chitinous exoskeleton is present around their body.
- **Reproduction:** These animals are unisexual.
- **Examples:** Crab, spider, scorpion, millipede, centipede, cockroach, butterfly, honeybee, etc.

8) **Phylum- Mollusca**

- **Body Structure:** Body of these animals is soft and slimy. Hence, they are referred as mollusc.
- This is second largest phylum in animal kingdom.



- **Habitat:** These animals are aquatic or terrestrial. Most of the aquatic molluscs are marine, but few are freshwater dwellers too.
- **Germinal Layers coelom:** Body of these animals is triploblastic, eucoelomate, nonsegmented and soft.
- **Body Symmetry:** Except animals like snail, their body shows bilateral symmetry. Their body is divided into three divisions like head, foot and visceral mass.
- **Characteristics:** Visceral mass is covered with mantle. This mantle secretes a hard, calcareous shell. This shell may be external or internal or even absent in some cases.
- **Reproduction:** These animals are unisexual.
- **Examples:** Bivalve, Snail, Octopus, etc.

9) **Phylum- Echinodermata**

- **Body Structure:** Calcareous spines are present on the body of these animals; hence they are called as echinoderms.
- **Habitat:** These animals are found only in ocean.
- **Germinal Layers coelom:** Their body is triploblastic, eucoelomate.
- **Body Symmetry:** It is radially symmetrical in adult stage. However, they show bilateral symmetry in larval stage.
- **Locomotion:** They perform locomotion with the help of tube-feet. Tube feet are also useful for capturing the prey. Some animals are sedentary.
- **Characteristics:** They have skeleton made up of calcareous spines and / or ossicles (plates).
- **Reproduction:** These animals have good ability of regeneration. These animals are mostly unisexual.
- **Examples:** Star fish, sea-urchin, brittle star, sea-cucumber, etc.

10) **Phylum-Hemichordata**

- **Body Structure:** Body of these animals is divided into three parts as proboscis, collar and trunk.
- Notochord is present in proboscis region only. Hence, they are called as hemichordates.
- These animals are also called as 'acorn worms'.
- **Habitat:** These are marine animals, live in burrows in sand.
- **Characteristics:** They have one to many pharyngeal gill slits.
- **Reproduction:** They are unisexual, or some may be hermaphrodite.
- **Example:** Balanoglossus, Saccoglossus.

Chordates:

Phylum-Chordata

Depending on the notochord, the Phylum chordata is classified into three subphyla.

A. Subphylum - Urochordata

- Urochordates are animals having notochord only in tail region of larvae.
- Marine, sedentary adults.
- Larvae are free swimming; they metamorphose into sedentary adults.
- Body is covered by test or tunic. This is a covering like a skin.
- Hermaphrodite animals.
- Examples: Herdmania, Doliolum, Oikopleura, etc.

B. Subphylum - Cephalochordata

- Notochord is present in entire body length(head to tail).
- Small, fish-like, marine animals.
- Large pharynx having gill-slits.
- Unisexual animals.
- Example: Amphioxus.



C. Subphylum - Vertebrata/Craniata

Vertebrata has been divided into six classes.

1. Class- Cyclostomata:

- **Body Structure:** These animals have jaw-less mouth provided with sucker.
- Cold blooded animals
- **Appendages:** Paired appendages are absent.
- **Exoskeleton:** Soft skin without any scales. Endoskeleton is cartilaginous.
- **Habitat:** Most of the animals are ectoparasites. Marine, aquatic
- **Reproduction:** sexual, Oviparous-Eggs
- **Examples:** Petromyzon, Myxine, etc.

2. Class- Pisces

- **Habitat:** Aquatic animals living in marine and fresh waters.
- **Body Structure:** Body is spindle shaped to minimize water-resistance.
- Cold blooded animals
- **Appendages:** They have paired and un-paired fins for swimming. Tail fin is useful as a steering organ during swimming.
- **Skeleton:** Exoskeleton is in the form of scales and endoskeleton is either cartilaginous or bony.
- These are cold blooded (Poikilotherms)
- **Respiration:** Respiration occurs with gills.
- **Reproduction:** Oviparous-lay Eggs
- **Example:** Rohu, Pomfret, Sea horse, Shark, Electric ray, Sting ray, etc.

3. Class- Amphibia

- **Habitat:** These animals are strictly aquatic during larval life and perform only aquatic respiration whereas they can live in water as well as on land during adult life.
- **Appendages:** They have two pairs of appendages. Digits are without claws.
- **Skeleton:** Exoskeleton is absent. Skin is moist for respiration.
- **Body Structure:** Neck is absent. Eyes are prominent with eye lids. External ear is absent but tympanum is present.
- Cold blooded animals
- **Respiration:** During tadpole/larval stage by gills, In adulthood by skin in water and by lungs when on land
- **Reproduction:** Eggs, larva, adult metamorphosis.
- **Example:** Frog, Toad, Salamander, etc.

4. Class- Reptilia

- **Habitat:** These are first true terrestrial animals with creeping movement. Few are aquatic
- These are cold blooded (poikilotherms) animals.
- **Appendages:** Two weak pairs of limbs. Digits are provided with claws.
- They creep on the land as their body cannot be lifted up.
- **Skeleton:** Their skin is dry and scaly.
- **Body Structure:** Neck is present between head and trunk. External ear is absent (tympanum is present).
- **Reproduction:** Sexual, oviparous, young one, adults.
- **Examples:** Tortoise, Lizard, Snake, etc.

5. Class- Aves

- **Habitat:** These vertebrates are completely adapted for aerial life.
- These are warm blooded (Homeotherms).



- **Body Structure:** Their body is spindle-shaped to minimize air resistance during flight. Neck is present between head and trunk. Jaws are modified into beak.
- **Appendages:** Forelimbs are modified into wings. Digits are covered with scales and bear claws.
- **Exoskeleton:** Exoskeleton is present in the form of feathers.
- **Reproduction:** Oviparous, parental care.
- **Examples:** Peacock, Parrot, Pigeon, Duck, Penguin, etc

6. Class- Mammalia

- **Habitat:** Mainly terrestrial, few aquatics, few aerials.
- Presence of mammary glands is typical character of mammalia.
- These animals are warm blooded. (Homeotherms).
- **Body Structure:** Body is divided into head, neck, trunk and tail.
- **Appendages:** Digits are provided with nails, claws, or hooves.
- **Skeleton:** Exoskeleton is in the form of hairs or fur.
- **Respiration:** By lungs aided with diaphragm.
- **Reproduction:** Few Oviparous, majority viviparous. Birth to live young, mammary glands secreting milk.
- **Examples:** Elephant, Human, Kangaroo, Dolphin, Bat, etc





Class - X

2023-24

Board: Maharashtra

Introduction to Microbiology

- **Applied Microbiology:** Study of the enzymes related to some prokaryotes and eukaryotic microbes, proteins, applied genetics, molecular biology, etc. is done in the applied microbiology.
- **Industrial microbiology:** This science is related to commercial use of microbes in which various economic, social and environment related processes and products are included.

PRODUCTS AND PROCESS:

□ **Dairy Products:**

- By preparing various milk products, one can preserve milk for a long time.
- Cheese, butter, cream, Kefir, yoghurt, etc. are some important milk products.

□ **Basic process for production of milk products:**

- Pasteurization of milk initially to eradicate other harmful bacteria.
- Later fermentation using Lactobacilli.
- Lactose sugar present in the milk > Converted into Lactic acid.
- Lactic acid coagulates the milk proteins.
- Compounds with taste and flavour are formed, such as Diacetyl having buttery flavour.

□ **Yoghurt production:**

Method of industrial production of yoghurt:

- Yoghurt is one of the milk products produced from milk with the help of lactobacilli (inoculant).
- In the industrial production of yoghurt, the milk is added with condensed milk powder. Then this milk is subjected to fermentation.
- Milk is boiled and then it is cooled till it becomes lukewarm.
- Then the bacterial strains of *Streptococcus thermophilus* and *Lactobacillus delbrueckii* are added to this lukewarm milk in 1:1 proportion.
- The Streptococcus bacteria convert the milk into solution containing lactic acid. This makes the proteins to gel out. It makes the yoghurt dense.
- The lactobacilli helps in the formation of acetaldehyde like compounds giving a characteristic taste to the yoghurt.
- For commercial reasons, various fruit juices can also be mixed with yoghurt to impart different flavours forming Strawberry yoghurt, banana yoghurt, etc.
- The pasteurization is carried out to Increase the shelf life of yoghurt and improve its probiotic properties.

□ **Butter:**

Butter is of two types: Sweet cream and cultured. For large scale production of cultured butter, microbes are used.

□ **Cheese production:**



Method for cheese production:

- (1) Initially, chemical and microbiological tests of milk done.
- (2) *Lactobacillus lactis*, *Lactobacillus cremoris*, and *Streptococcus thermophilus* colours added to the milk.
- (3) Milk turns sour > whey (water in yoghurt) is removed to make milk dense.
- (4) Protease enzyme from fungi is used at present to produce vegetarian cheese.
- (5) Previously, rennet enzyme, from alimentary canal of cattle was used.
- (6) Whey is completely separated from yoghurt.
- (7) To summarise, the steps of process of cheese formation: cutting the solid yoghurt into pieces → washing → rubbing → salting → mixing of essential microbes + pigments + flavours → Cheese is pressed and cut into pieces → stored for ripening.

□ Probiotics:

Milk products containing active bacteria such as *Lactobacillus acidophilus*, *Lactobacillus casei*, *Bifidobacterium bifidum*, etc. form probiotics. All the above microbes help in maintaining the intestinal bacteria which help the digestion. They also decrease the harmful bacteria such as *Clostridium*.

□ Benefits of probiotics:

- The colonies of useful bacteria are formed in the alimentary canal.
- The harmful microbes can be controlled by reducing their metabolic activities.
- Probiotics improve resistance to fight against diseases.
- Some harmful substances are formed during metabolic activities. Their bad effects are lowered due to probiotics.
- Due to antibiotic treatment, useful bacteria become inactive. Probiotics revitalize them by making them active again.
- Probiotics are used for treatment of diarrhea,
- In poultry treatment too, probiotics are used.

□ Bread:

Bread is made from flour obtained from cereals.

- The dough is made by mixing together, the baker's yeast *Saccharomyces cerevisiae*, water, salt and other necessary materials.
- The yeast helps in the fermentation and brings about the conversion of sugar into carbon dioxide, and ethanol.
- Due to released CO₂ the bread becomes spongy.
- Yeast is of two types: (a) Compressed yeast containing carbohydrates, fats, proteins, various vitamins, and minerals. This is used in commercial bakery industry. (b) Dry, granular yeast used for domestic use.
- Products produced with yeast become nutritive.
- Vinegar, soya sauce and monosodium glutamate (Ajinomoto) are also produced by microbial fermentation.

□ Vinegar:

Vinegar Production: Vinegar is 4% acetic acid CH₃COOH.

Uses of vinegar:

- Impart sour taste to food materials.
- Preservation of pickles, sauce, ketchup, chutneys, etc.



- By fermentation of fruit Juices, maple syrup, sugar molasses, starch of the roots, etc. with the help of yeast *saccharomyces cerevisiae* ethanol is obtained.

□ **Preparation of vinegar:**

Ethanol + *Acetobacter* and *Glucanobacter* mixture → microbial degradation → Acetic acid and other by-products → rarefaction of mixture → Separation of Acetic acid → Acetic acid bleached with potassium ferrocynide → Pasteurization → very small quantity of SO₂, gas is mixed → Vinegar is produced.

□ **Soya sauce:**

- Fungus *Aspergillus oryzae* is used to ferment the mixture of flour of wheat or rice and soyabean to produce soya sauce.

□ **Beverages:**

Microbial Enzymes: The enzymes obtained by the activity of microbes are used in the industrial processes. These microbial enzymes are better than the chemical catalysts.

- Microbial enzymes work at the low temperature, pH and pressure.
- Energy is saved in the processes using microbial enzymes.
- There is no need of erosion-proof instruments for the processes.
- Only specific processes are carried out by the enzymes.
- Unnecessary by-products are not formed during the reactions.
- Purification is not needed and hence expenses on purification are reduced.
- Elimination and decomposition of waste material is avoided.
- Enzymes can be reused.
- Microbial enzymes are thus eco-friendly.

□ **Examples and Use of Enzymes:**

Some examples of microbial enzymes are:

1. oxidoreductases
2. transferases
3. hydrolases
4. lyases
5. Isomerases
6. ligases

Use of microbial enzymes in Industries:

1. Detergents: Used in cleaning process to remove dirt at low temperature.
2. Obtaining glucose and fructose syrup from corn flour using enzymes of bacilli and Streptomyces.

□ **Amino acids:**

Amino acids: Some microbes are used in the production of amino acids. These amino acids are used in making commercial products. L-glutamic acid, citric acid, gluconic acid, lactic acid and itaconic acid are such amino acids which are prepared by action of certain microbes on some plant sources.

□ **Xanthan gum:**



- *Xanthomonas* species is used to make Xanthan gum by fermentation of starch and molasses.
- It is useful due to properties like solubility in hot and cold water and its high density.
- It is used in the production of pigments, fertilizers, weedicides, textile Pigments, toothpastes, high quality paper, etc.

□ Substances obtained by microbial processing are as follows:

- (1) To make the medium acidic: Citric Malic and Lactic acid
- (2) For binding the proteins: Glutamic acid, Lysine, Tryptophan
- (3) Microbial restrictor: Nycin and natamycin
- (4) Antioxidants, vitamins: Ascorbic acid (Vit. C), B₁₂, B₂
- (5) Edible colours: Beta carotene, lycopenes, xanthenes, lutein
- (6) Emulsifiers: Polysaccharides, glycolipids
- (7) Essence: Vanillin, Ethyl butyrate (fruit flavour), peppermint flavour, essence of various fruits and flowers
- (8) Artificial sweetener (low calorie): Xylitol, aspartame

□ Antibiotics:

Antibiotics: Carbon compounds obtained from some bacteria and fungi which are able to destroy or prevent the growth of harmful micro-organisms are called 'antibiotics'.

In medical field Antibiotics are medicines obtained from different types of bacteria and fungi by carrying out certain processes.

Some human diseases and veterinary diseases have been controlled due to antibiotic treatment.

Antibiotics which are used against Gram positive and Gram negative bacteria: (1) Penicillin, (2) Cephalosporins (3) Monobactam (4) Bacitracin (5) Erythromycin (6) Gentamycin (7) Neomycin (8) Streptomycin (9) Tetracyclines (10) Vancomycin, (11) Rifamycin - against tuberculosis.

□ Bio-fuel:

Biofuel is available in the following three forms:

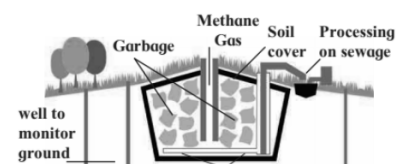
- Solid: Coal, dung, crop residue
- Liquid: Vegetable oils, alcohol
- Gaseous: Gobar gas and coal gas.

□ Land-filling sites:

- The type of degradable waste which is created in the urban areas is dumped in the land filling sites which are away from the residential areas.
- The large pits dug and then lined with plastic sheets are used for dumping compressed waste.
- It is covered with certain bioreactor substances and specific bio-chemicals.
- Microbes present therein decompose the biodegradable materials.
- The pit is sealed with soil slurry after it is full.
- Compost is formed by decomposition after few days. This is removed and such land filling sites can be reused after removal of compost.

□ Sewage Management:

- In cities, the sewage is sent to processing plant and is treated with microbes.





- Microbes that carry out decomposition, are mixed with sewage. Such microbes are able to destroy pathogens as well as decompose any compounds.
- Upon decomposition of the carbon compounds present in sewage, microbes release methane and CO_2 . Xenobiotic chemicals are present in sewage. The phenol oxidizing bacteria decompose such chemicals.
- The sludge formed in this process, is used as fertilizer.
- Environmentally safe water is released after microbial treatment done for the sewage water.

□ **Microbes and Functions in cleaning:**

- **Some microbes:** Remove the sulphur from fuels.
- **Thiobacilli and Sulphobacilli:** Conversion of heavy metals into compounds before leaching
- **Hydrocarbonoclastic bacteria (HCB):** *Pseudomonas spp.* And *Alcanovorax borkumensis*: To treat oil spills by destroying the pyridines and other chemicals. Conversion of hydrocarbons into CO_2 and water.
- ***Vibrio*, *Ideonella sakaiensis*:** Decomposition of PET (Polyethylene Terephthalate Polyester)
- *Actinomyces*, *Streptomyces*, *Nocardia*, *Actinoplanes* (all fungi): Decomposition of rubber from garbage.
- ***Acidophillium spp.* And *Acidobacillus ferrooxidans*:** Control of soil a pollution caused by acid rain.
- ***Geobacter*:** Conversion of salts of uranium into insoluble salts.

□ **Microbial Inoculants:**

- Microbes-containing inoculants are produced by process of fermentation.
- They are used for spraying on seeds before sowing. Some of them are released into plants.

□ **Benefits of microbial inoculants:**

- The nutrients are supplied through the microbial inoculants which help in growth of plants.
- The nutritional quality of crops is increased.
- Soil pollution caused by chemical fertilizers is prevented by injecting solution containing *Azotobacter* and artificial nitrogenase. Such methods are adopted in organic farming.
- Chemicals like fluoroacetamide are mixed with soil as due to use of chemical pesticides in agriculture. Such harmful substances can be eradicated by using microbes.

□ **Bioinsecticides:**

- Through biotechnology, bacterial and fungal toxins are directly integrated into plants. These plants are thus protected against fungal and pest attacks.



Cell Biology and Biotechnology

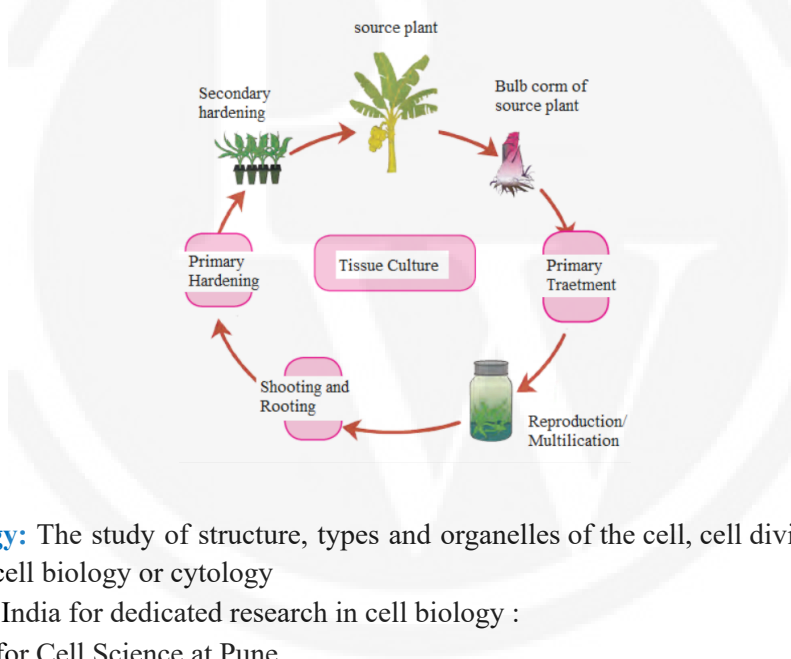
Cell: The structural and functional unit of the body is called a cell.

Tissue and functions of tissue: Tissue is a group of cells that perform a similar function. Every tissue performs a definite function. E.g.

- The muscular tissues in the body perform contraction and extensions thereby helping in locomotion.
- The conducting tissues of the plants like xylem and phloem transport the water and food respectively.

Tissue Culture:

Tissue Culture: Tissue culture is the technique in which 'ex vivo growth of cells or tissues in an aseptic and nutrient-rich medium' is done.



Cell Biology/Cytology: The study of structure, types and organelles of the cell, cell division and some other aspects together is known as cell biology or cytology

Research institutes in India for dedicated research in cell biology :

- (1) National Centre for Cell Science at Pune
- (2) Instem at Bengaluru

Stem Cells : These are special types of cells present in the body of multicellular organisms that can give rise to all other types of cells present in the body of multicellular organisms are called stem cells :

- The zygote formed after fertilization of gametes contains mass of cells, which are similar cells. These are the stem cells. They can give rise to new organism.
- Stem cells can be used to produce new tissues in laboratory.
- Stem cells also play an important role in wound healing.

Differentiation of stem cells :

- The formation of different types of tissues to perform specific and different functions is called differentiation of the stem cells.
- Once the new tissues are formed their capacity to differentiate is lost. Later they form cells which are similar to them.

Locations of stem cells where they found:



- In the umbilical cord present in the uterus of mother.
- In the blastocyst stage of embryonic development.
- In red bone marrow
- In adipose connective tissue of adult human beings.
- In blood

□ **Stem cell research :**

- (1) The revolutionary event that occurred in biotechnology is stem cell research.
- (2) Stem cell research will cause the fundamental changes in the medical science.
- (3) Stem cells are of two types depending upon the source, viz.
 - Embryonic stem cells
 - Adult stem cells.

□ **Embryonic stem cells :**

Embryonic stem cells :

- (1) Embryo is formed by divisions of zygote. These embryonic cells continuously divide by mitosis. These cells are stem cells.
- (2) But by 14th day of formation cell differentiation starts. This differentiation causes formation of 220 different types of cells, e.g. osteocytes (bone cells), hepatocytes (liver cells), and neurons, etc. from the stem cells.
- (3) Before the differentiation, the embryonic cells are called embryonic stem cells. These stem cells are primary, undifferentiated cells having ability to multiply. They are called parent cells of all types of human cells.
- (4) The property to develop into different types of cells is called pluripotency. This property is present in the stem cells.
- (5) Stem cells can be collected well before the beginning of differentiation and cultured with certain biochemical stimuli in laboratory. Based on the type of stimulus, stem cells transform themselves into desired type of cells, later the tissues and organs.

□ **Adult stem cells :** Stem cells taken from the adult body are called adult stem cells.

For adult stem cells following sources are used :

- Red bone marrow
- Adipose connective tissue
- Blood
- Cord blood can also be used immediately after birth
- Placenta is also rich source of stem cells.

□ **Uses of Stem Cells:**

1 In Regenerative Therapy:

- Cell therapy: Replacing the dead cells in patients of diabetes, myocardial infarction, Alzheimer's disease, Parkinson's disease, etc.
- The formation of blood cells in patients of anaemia, thalassemia, leukaemia, etc.

2 In Organ Transplantation:

- In organ-failure such as kidney and liver, some tissues can be formed by stem cells and transplanted in a needy patient.

□ **Organ transplantation :**

- (1) If less efficient or completely functionless organ of the body is changed, the life of such patient can be saved. Kidney transplantation or skin transplantation can be done on the patients if there is suitable donor for the same.



Investigations about the factors such as blood group, diseases, disorders, age, etc. are done for donor and recipient. This matching is done before the transplantation.

- (2) For kidney and skin, live donor can be taken but liver, heart, eyes can be donated posthumously, i.e. after death.

Biotechnology:

The techniques of bringing about improvements in living organisms by artificial genetic changes and by hybridization for the welfare of human beings is called biotechnology.

☐ **Use of biotechnology:**

- Biotechnology can be used in the fields of agriculture, horticulture, medical field, diagnosis of diseases etc.
- Biotechnology is used in the production of cash crops, improvement in varieties of cash crops, increase in abilities of plants to withstand environmental stresses,
- Biotechnology is used in vaccine production, early diagnosis of congenital diseases, organ transplant, cancer research, production of artificial skin, cartilage, etc.
- In laboratories.
- It includes the techniques of genetic engineering and tissue culture.

☐ **Impact of biotechnology on agriculture and other related fields:**

- (1) GMO or Genetically Modified Organisms are created. For this purpose, the DNA is changed in the crops. The modified crops then possess required features.
- (2) Following changes have occurred in the crops due to biotechnology.
 - The varieties that can withstand the changes in the environment can be created.
 - g., constantly changing temperature, dry and wet droughts, climate change and some similar stressor-resistant varieties are produced through genetic engineering.
- (3) GM crops are resistant to insect pests, pathogens, chemical weedicides, etc., due to their changed genome. This reduces the use and the cost of chemical pesticides which are harmful.
- (4) Due to use of seeds of GM crops, there is improvement in nutritive value of crops. Therefore now, seeds of a better-quality crops are being produced.
- (5) There is also a decrease in loss of crops. Subsequently there is increase in the cultivable land and agriculture.
- (6) Some insect resistant crops such as Bt Cotton have been produced which is widely used in Maharashtra.

☐ **Biotechnology and its Applications :**

- Cytology, biochemistry, molecular biology, and genetic engineering are different branches of science that are included in biotechnology.
- In the field of agriculture and pharmacy major progress has been done due to biotechnology.
- Improving the agricultural yield, production of high-class varieties of crops and suitable farm products through tissue culture are the aspects in which biotechnology is largely used.
- Pharmaceutical experiments to produce antibodies, vitamins, and hormones like insulin are taking support of biotechnological principles.

☐ **Benefits of Biotechnology :**

- (1) Increasing the per hectare yield of the crop land, irrespective of the limitations.
- (2) By development of resistant varieties, the expenses on disease control become less.
- (3) Fast fruit setting varieties are developed that increase the per annum yield.
- (4) Varieties which can withstand changing environmental parameters such as variable temperature, water-stress, changing fertility of soil, etc. are created. These are stress resistant varieties.



□ **Commercial Applications of Biotechnology:**

Crop Biotechnology: Biotechnology is used in agricultural field to improve yield and variety.

- Hybrid Seeds: This is especially useful for fruits.
- Genetically Modified Crops: High yielding varieties with resistance to diseases, alkalinity, weeds other stresses like cold and drought.
- Bt Cotton: The toxin which is fatal for bollworm was produced in leaves and bolls of cotton.
- Bt Brinjal: This improved variety of brinjal kills the pest in same way as the Bt cotton does.
- Golden Rice: A gene synthesizing the vitamin A (Beta carotene) has been introduced in this variety of rice, contains 23 times more amount of beta carotene than normal variety.
- Herbicide tolerant plants: Due to this, it has become possible to selectively destroy the weeds.

□ **Biofertilizers:** Bacteria like Rhizobium, Azotobacter, Nostoc, Anabaena and plants like Azolla are used as biofertilizers

□ **Animal Husbandry:**

- (1) In animal husbandry, there are two main methods used: (a) Artificial insemination (b) Embryo transfer
- (2) By these methods, the quantity and quality of animal products such as milk, meat and wool are both improved.
- (3) Animal varieties that have more strength have been developed for hard work.

□ **Human Health:**

- Biotechnology for human health management takes into consideration two important tasks, these are diagnosis and treatment of the diseases.
- Identifying the role of gene, in disease of a person. Diagnosis of diabetes and heart diseases can now be done even before the onset of symptoms.
- AIDS and dengue can be diagnosed within few minutes due to biotechnology. Treatment in such cases can be done at the earliest.
- Various vaccines and antibiotics are also made in this way.

Vaccines and Vaccination: Vaccine is the 'antigen' containing material given to acquire either permanent or temporary immunity against a specific pathogen or disease.

- Initially the completely killed or partially killed pathogen is used in the vaccine
- But now a days the antigen present on the pathogen body is synthesis with the help of biotechnology and it gives to the people.
- Transgenic potato is produce by biotechnology and it contains antigen.

Interferon: This is a group of small sized protein molecule used in treatment of viral diseases.

Gene therapy: Gene therapy to treat genetic disorders in somatic cells has become possible due to biotechnology. Ex. Phenylketonuria (PKT)

Cloning: Cloning is the modern technique in which there is production of replica of any cell or organ or entire organism is done. There are two types of cloning, viz.

(i) Reproductive cloning and (ii) Therapeutic cloning.

Reproductive cloning: In reproductive cloning, a clone is produced by fusion of a nucleus of diploid somatic cell with the enucleated ovum of anybody. In the process, the sperm or male gamete is not needed.

Therapeutic cloning: This technique is largely used for treatment purpose. Stem cells are derived from the cell formed in laboratory by the union of somatic cell nucleus with the enucleated egg cell.

- Cloning technique is used for therapy of various diseases.
- Gene cloning can also be done to form millions of copies of same gene. Such genes are used for gene therapy and other purposes.



- Due to cloning technique, the inheritance of hereditary diseases can be controlled, continuation of generations can be achieved, and certain characteristic genes can be enhanced.
- However, for human cloning, there is worldwide opposition due to ethical reasons.

□ **Environment and Biotechnology:**

- (1) For treatment on sewage and solid wastes, microbial processes are used to speed up decomposition. If such organically rich sewage is released in natural water bodies, there is oxygen depletion in water. This is because the organic matter is oxidized with the help of dissolved oxygen in the water. Depletion in oxygen level adversely affects the aquatic life. Therefore, sewage is first oxidized by microbial techniques and then it is released.
- (2) Solid organic waste material is treated with microbes to produce compost on a large scale.
- (3) Some new concepts in biotechnological methods are bio-remediation, biopesticides, biofertilizers, biosensors, etc.
- (4) In the process of bioremediation absorption or destruction of toxic chemicals and harmful pollutants is carried out with the help of plants and microorganisms. Phytoremediation is the method of bioremediation in which plants are used for this process.

□ **Examples of Phytoremediation:**

- The *Pseudomonas* bacteria are useful for cleaning the hydrocarbon and oil pollutants from soil and water.
- The fern *Pteris vitata* can absorb the arsenic from the soil.
- Genetically modified variety of Indian mustard can absorb selenium from soil.
- Sunflower can absorb uranium and arsenic.
- The bacterium *Deinococcus radiodurans* is highly radiation resistant organism. It has been genetically modified and used to absorb the radiations from radioactive debris.
- Grasses like alfalfa, clover and rye are used in phyto-remediation.

DNA fingerprinting :

- (1) In every organism, the nucleotide sequence in the DNA molecule is unique. In human beings as a fingerprint is unique for every person, similarly DNA is also unique. Therefore, the technique to identify any person can be proved by his or her DNA.
- (2) DNA fingerprinting technique is mainly used in
 - Investigations of forensic sciences.
 - To establish identity of the criminal.
 - To identify parents of any child in case of disputed parentage.
- (3) In India at "Centre for DNA fingerprinting and Diagnostics", located at Hyderabad DNA fingerprinting is done.

□ **Important stages in Agricultural Development :**

Green revolution: Dr. Norman Borlaug (USA) and Dr. M. S. Swaminathan (India) have valuable contribution in green revolution.

- (1) The improved methods for harvesting maximum yield from minimum land are collectively called green revolution.
- (2) In green revolution, methods of improvised dwarf varieties of wheat and rice, proper utilization of fertilizers and pesticides and appropriate water management have led to increased production of food grains.
- (3) New varieties of various crops have been developed through different agricultural research institutes.

□ **White revolution:**

- (1) The cooperative dairy movement that was undertaken with the help of biotechnology to produce abundant milk and dairy production, is called white revolution.



- (2) At Anand in the State of Gujarat, the cooperative movement was laid by Dr. Verghese Kurien. He took this movement to the greatest heights. AMUL (Anand Milk Union Limited) which was founded by him, has now become an international brand.
- (3) The self-sufficiency in dairy business was achieved due to white revolution. Many experiments done for quality control, newer dairy products and their preservation.

□ **Blue revolution:**

- (1) Blue revolution is production of various useful aquatic organisms with the help of water.
- (2) In East Asian region, fish species are raised in the farm ponds. This practice is common there. This is known as aquaculture. Rearing fish is called pisciculture while culturing other aquatic organisms is aquaculture, that of marine organisms is mariculture.
- (3) Government of India encourages the aquaculture activities. Government has launched the program 'Nil-Kranti Mission-2016' (NKM-16). Subsidies at 50% to 100% are offered in this case.
- (4) Aquaculture and mariculture can be carried along Indian coastline on a large scale. Fresh water fishes like rohu, catla and marine and brackish water prawns and lobster, etc. can be cultured on large scale.

□ **Fertilizers:** Two types of fertilizers are used in agriculture. One of those is organic manure and others are chemical fertilizers.

□ **Organic (Manure):**

- Improved water holding capacity of the soil. Soil conservation takes place.
- Due to humus formation, upper layer of the soil is formed.
- Essential elements (N, P, K) are easily available due to activities of earthworms and fungi.

□ **Chemical fertilizers:**

- Decrease in fertility of soil due to excessive use.
- Toxic for soil bacteria, cattle, human.
- Contamination of the environment.

□ **Insecticides:**

- Plants have natural immunity to fight against infections.
- To eradicate the pests, insecticides are used on plants. This use can be harmful for the environment.
- Natural pest control measures such as insectivorous frogs and birds keep the pest population under control.
- But in an attempt to increase the yield, man excessively uses the pesticides and insecticides.
- These are toxic to all resident organisms. They also show biomagnification, in which the concentration of the insecticide goes on increasing according to the food chain.
- The insecticides like DDT, malathion, chloropyrifos also contaminate water and soil.

□ **Organic farming:**

- When the chemical fertilizers and pesticides are not used, only local, sturdy varieties are cultivated, and a natural balance is maintained, then such type of farming is called organic farming.
- Organic farming and organic products are in demand these days.
- The use of chemical fertilizers and pesticides cause many environmental problems due to pollution. The soil fertility is decreased, the insect-pests become resistant and cause more damage.
- Organic farming can help to save the environment from such effects.

□ **Apiculture:**

- (1) Apiculture is rearing of honeybees to obtain products such as honey and wax.
- (2) In such practice, artificial hives are placed. By such practice, the honey can be extracted without causing any harm to the honeybees or their hives.



□ **Cultivation of Medicinal Plants :**

- (1) In Ayurveda, different medicines are traditionally obtained from plants. The rich biodiversity of India was conducive in gathering such medicinal herbs from forests.
- (2) Now due to deforestation, the medicinal plants and herbs are becoming rare. Hence their cultivation is to be done.

□ **Fruit Processing:**

- (1) Fruit processing can form following products: Chocolates, juices, jams and jellies. Since fruits are perishable agro-produce, processing is a must.
- (2) Different methods of fruit processing are depositing in cold storage, drying, salting, airtight packing, preparing jams (muramba), condensing, etc.





Class - X

2023-24

Board: Maharashtra

Social Health

- **Social Health:** The ability of a person to establish relationship with other persons is called social health. One of the important characteristics of social health is the ability to change one's own behaviour according to changing social conditions.
- **Factors affecting the social health:**
 - Satisfying basic needs of people - food, clothing, shelter, medicines
 - Residential areas, Transport facilities, Playgrounds, Gardens, Water, Toilets
 - Education and job opportunities, financial status
 - Social security, Social treatment, Social environment of the surroundings.
 - Social and Physical conditions of the surrounding
 - Political conditions (views).
- **The factors necessary for good social health:**
 - Strong personality, having large number of friends and relatives.
 - Proper time management, trust in others, respect and acceptance for others.
- **Factors disturbing or endangering the social health:**
 - 1. Mental Stress:**
 - Due to population explosion, there is increased competition in every aspect such as education, employment and business.
 - Due to nuclear families and working parents, children are facing the problems of loneliness and mental stress. Parents and especially mother being away from home has resulted into stress in children.
 - There is unequal treatment given to girl and a boy from the same house. The discrimination between brother and sister causes mental stress to girls due to such gender inequality.
 - In community, the adolescent girls are faced with the problems like teasing and molestation.
 - All the people are facing the stress due to ever increasing disorder, crime and violence.
 - People are in the quest of getting 'fast and easy money' become part of the system of crime. This is a social illness which is very hazardous effect of changing attitudes of people.
 - 2. Addiction:**
 - In adolescent age group, there is tremendous pressure of peers. This peer-group influence can be at times wrong, if the friends are not good. Instead of following advice of parents, the adolescent girls and boys tend to listen to the wrong advices of their friends.
 - Due to lack of parental supervision, children in their early age start using tobacco, cigarette, gutkha, alcoholic drinks, drugs, etc. This may be due to peer-pressure.
 - The children fall into the trap of addictions either due to peer-group pressure or due to false symbol of high standard living. Sometimes they try to imitate their elders.
 - The addictive substances are hazardous, and they cause long term effects. Some are temporarily intoxicating substances obtained from the plants. While some of the chemical ingredients in them can permanently damage the human nervous system, muscular system, heart, etc. Some tobacco like substances are carcinogenic in action especially on the mouth and lungs.



3. Incurable disease:

- Incurable diseases such as AIDS, TB, leprosy can spoil the social health.
- Care of patients of mental disorders and that of the old persons is very crucial to keep the social health in proper order.
- Due to changing pattern of our life style, there is increase in old age homes. In order to maintain the social health, we have to care for the older people in our community.
- Similarly, mentally disturbed patients should be given love and care along with prompt medical help.

4. Communication Media and excessive use of Modern Technology:

- Modern technology and communication media have drastically Changed the lives of people. Mobile phone has become an addiction that can endanger the social health. People remain constantly on the phones.
- Due to addiction to electronic media, various physical problems like tiredness, headache, insomnia (inability to sleep) forgetfulness, tinnitus (ringing or buzzing in the ears), Joint pains and problems in vision are on rise, This may be due to radiation of cell phones,
- Such radiations penetrate the bones of children causing damage, Person who are on computers and internet continuously become solitary.
- Negative computer and cell phone games cause children to become negative in their thoughts and actions.
- The internet has such huge information that it can provide information for every question. But its use should be done for only positive purposes.

□ **The symptoms of mental ill health:**

- Mentally ill persons are those who are indulging in such destructive and unnatural activities.
- Selficide persons constantly take selfies as they are not aware about their surroundings and the risks that may make them victims. Person indulging in selfie is not aware about the world around and the risks.
- Persons who commit domestic violence.
- Suicidal thoughts, sending messages to others and then committing suicides. Such sick people actually need support and they want others to sympathise with them.
- Spending lots of time on television, computer or cell phone without having conscience about time management is also a sign of mental illness.

□ **Cybercrimes:** Misuse of digital media is a cybercrime. Following mentioned incidences are examples of cyber crime

- No personal information should be shared on the phone, especially the details of bank account, Aadhaar card, PAN card, credit card or debit card number etc. Cheating persons by using this information is a greatest cyber-crime.
- If PIN of any debit or credit card is known to a stranger, he or she can make fraudulent transactions.
- The PIN number and CVV number should be kept total secret.
- In on-line purchases, many times consumers are cheated. In this, the consumers are shown superior items on websites but actually the inferior ones are sold to them.
- 'Hacking of information' is done by some programmers in which the confidential information about government, institutes and companies is obtained from internet with the help of computer programs.
- Fake Facebook accounts are opened, and false information is displayed there. This is for harassing girls or financially exploiting others.
- In internet piracy, written literature, software, photos, videos, music, etc. of other persons are misused or illegally used.
- Misuse of electronic media sending derogatory messages, spreading vulgar pictures and provocative statements is also a cybercrime.

□ **Cybercrime unit:** This is the department that collects the details and investigate the cybercrime and thereby find the criminal with the help of the internet. This department is under the police department. It has all experts of the field of internet technology.



□ **IT Act-2000:**

- The IT Act is enacted since 17 th October 2000. It was amended in 2008.
- Cybercriminal gets the imprisonment for 3 years or fine up to 5 lakh as a punishment.
- Maharashtra state is the first state to start cybercrime unit which is controlling the cybercrimes.

□ **Stress management:**

Ways to manage the stress:

Laughter club: People gather together and relieve their mental stress by laughing out loudly. It is thought that by laughing together, the stress is reduced.

- Communicating with all the people who are significant in our life especially with our parents.
- Expressing our feeling with near and dear ones and noting down our feelings.
- Hobbies like collecting curios, photography, reading, cooking, sculpturing, drawing, rangoli, dancing, etc. can relieve the stress in a major way.
- Diverting the energy and mind towards the positive thinking and removing negative thoughts needs to be practiced for a stress-free life.
- Learning and listening to the music and singing reduces the stress.
- Physical and outdoor games, exercise, discipline.
- Yoga, meditation, deep breathing, yogic sleep techniques.
- Massaging, visiting the spa too help to relieve the stress.
- Balanced and good food, and meditation,
- Enjoying nature, gardening, bird watching, lingering in nature, rearing a pet.

Mental problems:

(1) Depression

(2) Frustration

Nowadays, there is fierce competition, insecurity and criminal tendencies in the society. This kind of atmosphere is increasing mental and emotional stress. If the stress remains buried in the mind, persons are depressed or frustrated. This causes, mental disorders if not treated in time. Depression can lead to addictions. The suicidal thoughts hover in the mind.

Example: Girls are facing the problems of stress due to gender in equality.

Treatment:

- (1) Medical guidance and help
- (2) Counselling
- (3) Psychotherapy
- (4) Help to be taken from NGO.

Importance of outdoor games: Outdoor games give good physical exercise.

These games give many physical benefits. It improves personal discipline, interaction with fellow players and created sense of unity. Through play by driving away the loneliness, mental stress and depression is reduced. The person becomes more social. Therefore, it is said that the importance of outdoor games is unparalleled.

Importance of good communication: If the stress remains buried in the mind, persons are depressed or frustrated. Depression can lead to addictions. The suicidal thoughts hover in the mind. If at that phase we can open our heart by good communication, many problems can be solved.

- Help from counsellors can be taken to relieve the stress.
- By good communication with parents or family members harmonious relations can be re-established



Importance of exercise, yoga and meditation:

- Exercise, yoga and meditation are the ways to reduce mental and physical stress.
- In yoga various asanas and pranayama are performed. It also includes good food and discipline of the body and mind.
- Deep breathing, yog-nedra can help in the building up health.
- Meditation helps in concentration and brings positivity to the mind. Especially, the students increase the concentration in the studies.

□ **Effects of alcoholism:** Alcoholism is the addiction to have alcohol in the form of different types of liquor. Liquor is produced from alcohol. Alcohol is in turn obtained by fermentation of different substances.

- Consuming liquor becomes an addiction for a long-term. Due to alcohol, the efficiency of nervous system and especially the brain is affected.
- Other vital organs such as kidneys and liver are adversely affected.
- Lifespan of an alcoholic decreases due to constant drinking and malnourishment.
- Especially in adolescent age if alcohol is consumed the brain functioning does not take place properly. The mental ability of memorization and learning becomes slow. There is a lack of concentration in studies.
- The alcoholic person lacks the rational thinking and hence faces with social, mental and familial problems along with physical illness.
- If alcohol is manufactured in a wrong way by using any substrates, it forms the poisonous chemicals. Such chemicals can be lethal too.

□ **Institutional help:**

By many ways we all manage our stress. However, if those are not successful due to certain reasons, it leads to some more serious problems like depression and frustration. Medical advice, counselling and psychotherapy are available for such persons. Similarly, many NGOs also provide helping hand.

□ **Institutions that offer help:**

- **Unified Movement against Tobacco:** 45 different well-known organizations such as WHO, Tata trust, etc. have started the movement to control the tobacco consumption. The guidance is also provided to the activists who are active in anti-tobacco campaigns.
- **Salaam Mumbai Foundation:** This organization runs programs in various schools in Mumbai. The children from slum area are given opportunities in the field of education, sports, arts and business. The organization also helps the children in field of education and helps them to improve their health and lifestyle. This organization helps each school to take the oath of freedom from tobacco.
- **Government Schemes:** Phone number / helpline numbers are published in newspaper to help the children. Children in distress or facing any type of trouble can call on the helpline set by the Government. Proper help and guidelines are offered to children if they narrate the problem on this helpline.





Class - X

2023-24

Board: Maharashtra

Disaster Management

- **Disaster:** Disaster is the incidence that occurs suddenly causing heavy damage to life and property. The disaster can be manmade or due to natural reason.

OR

Hazardous event that suddenly take place in the surroundings is called disaster.

- According to the definition of disaster as given by United Nations: The sudden event that leads to the huge loss of life and property is called disaster.
- **Natural disasters:** Floods, wet and dry famine, cyclones, earthquakes, volcanoes, etc. are caused due to sudden change in the environment causing great damage. Since these disasters are due to nature's wrath, they are called natural disasters.
- **Man-made disasters:** When mankind needs natural resources for developmental purpose, environment gets

Classification of disasters:

Disasters can be classified depending on following main criteria.

- The extent of the damage caused.
- The period of disaster.
- The long term effects.
- The reason for the disaster.

There are two main types of disasters, natural and man-made. This is the preliminary Classification of disasters.

- **Second method of Classifying disasters:**

- Catastrophic disasters - Cyclones in Odisha, catastrophic earthquakes of Gujarat and Latur, frequently buzzing cyclones in coastal Andhra Pradesh, etc.
- Disasters making the impact for long duration: Famine, various problems of crops, strikes of workers, rising levels of oceans, desertification etc.

- **Types of disasters:** There are three types of disasters, viz. geophysical, biological and man-made.

Geophysical:

- Geological: Earthquake, volcano, tsunami, landslides, landfall, erosion, alkalization, flooding, etc.
- Climatic: E.g. Hot and cold waves, snow storms, snowfall, cyclones, hailstorm, drought, flood, meteorite, sunspots, etc.

Biological:

- Animal-origin: Infectious viruses, bacteria (cholera, malaria, hepatitis, plague, bird flew), insects, bite of venomous animals, etc.
- Plant-origin: E.g. Forest fire, fungal disease spreading (Blister), weed, (aquatic, carrot grass, common grass)

Man-made:

- Accidental or Unintentional: Toxic gases, Atomic test, Unplanned action, Accident
- Intentional: War, fire, bomb blast, forced migration, terrorism, rapes, child labour.



□ **The scope of the disaster is dependent on the following facts:**

- Pre-disaster phase
- Warning phase
- Emergency phase
- Rehabilitation phase
- Recovery phase
- Reconstruction phase

□ **Three sensitive phases of the disaster:**

1. Phase of emergency

- Maximum lives can be saved by quick actions in this phase.
- Actions such as search and rescue operations, medical assistance, first aid, restoring communication services, removing the people from affected area, etc. are done in this phase.
- Estimating the gravity of disaster is done in this phase only.

2. Transitional Phase

- In this phase the rehabilitation work starts.
- Clearing of debris, restoring water supply, repairing roads, etc. is done.
- Bringing normalcy in public life.
- Different institutes offer the monetary/ other help. This help is distributed to victims.
- Victims are offered with the permanent means of earning livelihood.
- Soothing the mental stress of victims.
- Rehabilitation of the victims is important aspect of this phase.

3. Reconstruction Phase

- Highly complicated phase.
- Begins in transitional phase. Reconstruction of buildings and roads and water supply facilities are restored.
- Restarting of farming practices.
- This phase takes long time for reconstruction.

□ **Disaster management:**

- Disaster management is the essential action that has to be undertaken by each nation and its citizens.
- Disaster management schemes should be dependent on every location, time and nature of the disaster.

The disasters are inevitable, but we have to overcome the hazardous after-effects of these. This fact has led to concept of disaster management.

- Effective disaster management is necessary in any kind of disaster. It may be minor or major, spread over long period of time or it may be short term.
- The disaster management should involve public participation. Disaster hits in a rapid way.
- In managing the after-effects of disasters, there should be arrangements to cope up with disastrous effects or there should be ways to make citizens capable to face and fight against the consequences.
- The losses should be minimized. Disasters are suddenly forcing us to deal with them, but by disaster management techniques, the losses can be minimized.

□ **Objectives of disaster management:**

- Saving human life from disasters. Helping them to move away from the place of disasters.
- Essential commodities to be supplied to the affected people. This will help to reduce the gravity of disaster.



- To bring back the conditions to normalcy.
- Rehabilitation of the affected and displaced victims.
- Protective measures for future in order to 'develop capability to face the disasters in future.

□ **Disaster management planning:**

- Through scientific and careful observations and analysis of previous data, the suitable action plans are developed for the disasters that might come in future.
- In such plans the preventive measures, rehabilitation and reconstruction is included. Such plans are kept ready for future.
- Disaster management is divided into pre-disaster management and post-disaster management.

□ **Pre-disaster management:** Before the disaster strikes, there are some measures taken to rescue people or to keep preparedness for the future calamities. This is called pre-disaster management.

- It includes following points:
- To identify the disaster prone areas.
- Through predictive intensity maps and hazard maps, the information should be collected about the probability of future disaster's intensity and probable sites of disasters, respectively.
- Learning by ourselves and then imparting special training to people to tackle with future disasters.
- Making general public aware about disaster management by training them through programmes. Spreading the message through mass media, etc.

□ **Post-disaster management:**

- Helping the victims of disasters in all possible ways.
- Local people who are saved from disaster should participate for further rescue work.
- Establishing the help centres for victims. Controlling centres, established are according to the type of disaster.
- Materials provided from the control centres for the victims are categorized, and distributed the same to victims with continuous review of the help.
- Preparedness for the rescue operations.

□ **Main aspects of disaster management cycle:**

- **Preparation:** If there is disaster in future, then a plan for minimizing the destruction is kept ready. This is called preparation for the disaster.
- **Preparedness:** In case of disaster, the general public and administration should immediately rise for action. This action is planned. This aspect is called preparedness.
- **Impact of disaster:** After the disaster recedes, its intensity is studied. How it has impacted is also analysed along with disaster management.
- **Response:** Just before the disaster and immediately after the disaster, the response given is to be analysed in this step.
- **Resurgence:** Resurgence forms an important link between emergency measures and national progress. It is a step showing extent of welfare and rehabilitation work done after the disaster. The progress of nation also becomes clear after analysing resurgence.
- **Redemption:** The damage is caused to the people of the nation; thus the planning is done to minimize such damage.
- **Restoration:** The return of the normalcy after the disaster is called restoration. The restoration time period for coming back to normal routine after the disaster may be different for different types of disasters.

□ **Functions of Disaster Management Authority:**

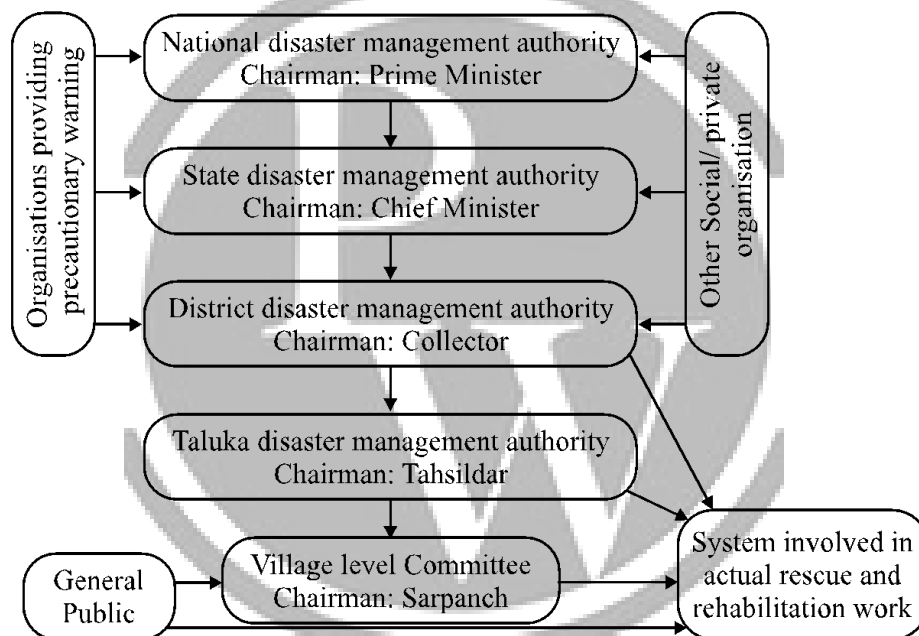


- Giving the necessary instructions.
- Assessing whether the instructions are followed or not.
- Reviewing the entire system.
- Controlling the actions.
- Planning and executing the plans.
- Designing schemes for each district, separately for each type of disaster and getting those sanctioned from state-level authorities.

□ **Structure of Disaster Management Authority:**

The Government has established an authority for management of disasters. Indian Government has also passed Disaster Management Act, 2005.

Control and coordination of disaster management is carried out from national level right up to village as shown in the following chart:



- **District Disaster Management Authority:** District collector is responsible for disaster management and implementation of rehabilitation schemes for that district. Planning, coordinating and controlling the implementation of rehabilitation programme is done by District Collector.

District-wise Disaster Control Unit:

- Establishment of district control unit is done immediately either after the impact of disaster or due to the warning of forthcoming disasters.
- Help and guidance from various agencies like army, air force, navy, telecommunication department, paramilitary forces, etc. is taken to review the various aspects of disaster.
- The control unit also coordinates with various voluntary organizations that help in disaster management.

□ **National Disaster Response Force:**

- The army jawans constitute the National Disaster Response force.
- It was established as per the Disaster Management Act, 2005.
- 12 divisions of this force work in the country.
- Its headquarter is located at Delhi and the army takes any required action throughout all the states of India.
- In Maharashtra, State Reserve Police Force carries out responsibilities of National Disaster Response Force.



- Jawans of this force help in rescue work at the times of disasters like cyclones, cliff-sliding, building collapse, etc.

First Aid and Emergency Action:

When suddenly any disaster takes place, the help given to the victims before the medical treatment is available, is called first aid. This knowledge of first aid becomes useful to save one's life.

- **First Aid Kit:** Material necessary for first aid is kept in the first aid kit. Such first aid kit Should be available at every place such as in schools, colleges, Malls, Company and offices, etc.

The necessary material in first aid box:

- For wrapping the wounds: Bandage strips of different size, Wound gauze, Triangular and circular bandage, Medicated cotton, Band aid
 - For cleanliness; Hand gloves, Clean and dry cloth pieces, Soap
 - Medicines: Antiseptic (Dettol/ Savlon), Petroleum jelly
 - Safety pins, Blade, Small pins, Needle, Scissors
 - Thermometer, Torch.
- **Mock Drill:** Mock drill is the activity which is carried out by creating virtual or apparent situation of disaster.
 - It is carried out to check the preparedness if the disaster approaches.
 - The responses of the people, the quickness of actions, etc. can be analysed prior to the actual disaster happens.
 - Trained personnel check execution of plan designed for disaster redressal. They are responsible for the training of people in case of disasters. The whole exercise helps to check the efficacy of the system prepared for disaster redressal.
 - Mock drill is organized in various schools to create the awareness among the school children in case of disasters. Such drills are taken with the help of fire fighters, by police force and by some voluntary organizations.
 - **Objectives of Mock Drill:**
 - Evaluating the response of the common people to the disaster.
 - Improving the coordination between various departments that work for the disaster control.
 - Identifying one's own abilities at the time of disaster.
 - Developing the ability to respond quickly to disaster.
 - Checking the competency of the planned actions.
 - Assessing the possible errors and risks.

